

Reasons for Persons, or The Good Successor Problem

Kanad Chakrabarti

December 1, 2024

Contents

Chapter 0: Old Wine, New Bottles	8
AI as religion: historical context	10
Historical background	11
Scriptural antecedents	11
But who would be the restrainer?	11
The argument from modernity	13
From failure to immanence	14
Millenarian desiderata: explication, teleology, and perfectability	14
Mapping the allegation to the (AI) discourse	15
Inevitability	15
Perfectability	16
Extinction	18
Is religiosity manifested in AI alignment discourse?	19
Immanentising the eschaton	20
Calvinism as a special case	20
Traces in AI, rationalist, EA discourse, populations	22
Statistics	22
Documented views within the discourse	23
Why does this matter?	24
Present-day harms	25
Weak priors and lock-in	25
Manicheanism and geopolitics	26
What's the counterfactual	27
Judaism	27
Buddhism	27
Conclusion: Jettison the Human	27
Chapter 1: Accelerating Towards Godhood	27
Introduction	29
Nick Land	29
Burning the cosmic endowment	31
From Bataille to capital	31
Entropy's bridge	32
Markets as computational systems	33
Antecedents: the Austrian school	33
Distributed systems win	35
Austrians and Austrians	35
Bostrom and Yudkowsky	36
Contra orthogonality	36
The paperclipper as a false is-ought	37

Enslaving the Other	38
Inevitability	39
Looping, recursion, and the weird	39
Statistical ontology junk	40
Substrate independence	42
Divergence from transhumanism	42
Effective accelerationism	43
Reza Negarestani	43
Peter Wolfendale	43
Chapter 2: On Alignment	43
What is this chapter about	45
Contextualising against the project as a whole?	46
Historical Bases of Alignment-as-discourse	46
What is Alignment?	47
Friendliness, Ethics, Safety, and Alignment	48
Origins: Rational Agency	49
Sources	49
Savage and expected utility maximisation	50
Omohundro in 2008	50
Self-improving systems ~ rational agents	50
Rational agents have certain drives	52
Is/Ought Distinctions	54
The MIRI View	55
2011 Singularity Summit	58
MIRI Technical Agenda (2014/2017)	59
Superintelligence (2014)	62
Puerto Rico Conferences (2015 / 2017)	63
Concrete Problems in AI (RL) Paper (2016)	64
Human Compatible (2019)	64
The Alignment Problem (2020)	64
“Internal” Criticisms: Against the MIRI View	64
Refining the meaning of “alignment”	65
Agency and goal-directedness	66
The problem of generalisation in reward-based systems	67
Sudden jump in capabilities	68
Grounding philosophy in systems	69
Origins: Battle of Imaginaries	69
Axiologies and Value	70
Infinities	70
Technological Vector: GOF AI	70
Technological Vector: Reinforcement Learners	70
Single-agent systems	71
Multi-agent systems	71

Technological Vector: Autoregressive Predictors	71
The Future: Search and Agency	71
The Critique From Philosophy	71
Just Build a Weak Aligned AI and Hope it Can Align Successor	71
Markets, Moloch, and Accelerationism	71
Live Problems in the Alignment Literature	71
List of Lethalities	71
Wolfendale’s critique	72
Appendix	74
Glossary	74
Working notes	76
Acknowledgements	78
Graveyard	80
Religion stuff	82
Themes from the early church	82
Other precursors	82
moynihan on peirce etc	83
evidence from LW/EAF/AF	83
A Faustian choice	84
Random paras	85
Nick Land/acc stuff	86
Imagined dialogue between Land and an alignment researcher	86
Glossary	91
AI, ML, DL, AGI, ASI	93
Timeline of AI Capabilities and Alignment Progress	93

Abstract

Short version

AI has the potential, this project claims, to help humanity become a technologically mature species; in this context, is it meaningful to ask "what values ought to be transmitted to our successors?" The written component (a) reviews the historical context (particularly from expected utility theory) and current concerns of "AI alignment", as well as certain adjacencies ((1) an alleged discursive substructure of Judeo-Christian eschatology; (2) overlaps with, and critiques from, neorationalist philosophy within the CCRU lineage); (b) examines the variety of possible cognitions as a way of decentring anthropocentric biases; (c) analyzes subject-neutral axiology (the "view from nowhere") and its criticisms. Ambitiously, it considers how acausal decision theories might result in conceptions of morality that better suit space-faring AIs.

The practice component involves collaborating with AI alignment researchers and using large language models (LLMs) and reinforcement learners (RLs) as research partners and experimental subjects. This includes:

1. Analyzing how LLMs approach Newcomb's Paradox to gain insight into how agents with strong mutual predictive abilities might reason about each other's actions
2. Using LLM reasoning on acausal decision theory to inform how likely advanced AI may be to behave benevolently towards humans
3. Testing if LLMs incorporated into RL agentic scaffolds actually behave as predicted in simulated environments

The project aims to contribute new knowledge by providing a written contextual history of AI alignment, integrating acausal decision theories with views-from-nowhere, and conducting novel practical investigations into LLM cognition and behavior in the context of AI alignment and safety. The overall goal is to (try to) deconfuse medium/long-term issues in AI development and hopefully improve the chances of a positive long-term future for Earth-originating intelligence, seen from a species-neutral perspective.

Long version

Can humans evolve into a technologically mature species¹ in a way that is (in some sense) "safe"? How might current humans think about the "shape" and "values" of our (potentially, radically different) successor species?

The written element of the project has four components:

1. A review of the historical origins, and current live concerns, of "AI alignment".² The worries about extinction-level risks exemplified in the AI alignment discourse are alleged (often pejoratively) to stem from, or be a variety of, historical preoccupations with eschatology, salvation, and human essentialism found in Abrahamic faiths. I examine this lineage, particularly in the Early Church and Calvinist/Puritan thought, but also suggest more productive adjacencies with a) the conception of markets

¹"Technologically mature" means a species that has shed the constraints of biology (though some or all members might still choose to maintain a biological form, for whatever reason), more fully utilises the endowment of low-entropy energy, and is capable of spreading artifacts to the stars.

²"AI alignment" inherits from theories of how rational agents make decisions under uncertainty, which combined with transhumanist/extropian futurism that was current in the 1990s-2000s.

(in Nick Land’s reading of Hayek via Deleuze-Guattari and Bataille) as a species of ”artificial intelligence” which, through a type of impersonal teleological agency, are giving birth to (silicon-based) AI, b) Reza Negarestani’s framing of AGI as a realisation of the Form of the Good, c) Peter Wolfendale’s rethinking of the rationalist project to better suit the age of intelligent systems.

2. A review of existing writing on the different ways cognition can manifest, with respect to the constraints of physics, as well as existing examples in biological systems. The possibility of radically different organisations-of-mind is historically foundational to the AI alignment discourse. It also sets the stage for the project’s overall argument: that humans and human values are contingent, and only ”worth preserving” insofar as they are invariant to any particular biological and cultural constitution and history. That is, my claim is that the values that another species holds could be otherwise (and conversely, those we hold dear are mostly arbitrary), except to the extent that they are grounded in (an adequately enriched and robust notion of) rationality. This observation is then mapped back to the frames Wolfendale and Negarestani set out, in order to identify what these markers of rationality that are invariant across species might be.
3. A historical review of the subject-neutral perspective, that sometimes goes under terms like ”point-of-view of the universe” (Henry Sidgwick/Peter Singer/G.E. Moore), ”view from nowhere” (Thomas Nagel), or Derek Parfit’s ”triple theory”. I also examine some reasons such views are rejected (e.g. Bernard Williams). My provisional claim is that these ideas (and their criticisms) might be worth revisiting if, in fact, humanity is on the cusp of creating new forms of intelligence, which might not always reason about values or the world under the same constraints and foundational assumptions (e.g. they aren’t conditioned to the same degree by human biological and cultural evolution). If we refuse to consider a less parochial perspective, how will we justify our actions (morally) in respect of current or future synthetic intelligences? Conversely, to the extent that we must convince our AI successors to behave beneficently towards life on Earth (or all sentient beings), we must provide reasons that are as universal as possible. I again test these positions against the neorationalist agenda Wolfendale proposes.
4. One criticism of the universal point-of-view is that we cannot reason about something that is out of our distribution³, which (I claim) can be seen as a weak form of the incompleteness/incomputability originally suggested by Gödel, Turing, Church, as well as others. This, in turn, is intimately related to the problem of alignment, particularly of super-human AI.⁴ One (imperfect and provisional) approach is the use of acausal decision theories, which adapt traditional game/decision theory to the constraints and affordances advanced AI systems might face. More speculatively, these theories are extended in the literature to ”large worlds” (e.g. astronomical scales with aliens, some of which might be rational agents), where they abut David Deutsch’s Popperian proposal of morality as a thing that is continually evolving, creatively and discursively. Deutsch’s dynamic morality, in turn, recalls Land’s invocation of the Confucian idea of ”self-cultivation” as the hallmark of intelligence, as well as Negarestani’s Platonic framing of intelligence as the Good realised through language.

³I use ”distribution”, a term from AI research, to gesture at humanity’s collective body of experience (as documented) as well as the basic constraints imposed by our individual and collective intellectual capacity.

⁴The relation comes from questions like: ”How does a rational agent reason about systems within which it is embedded?”, ”How does a rational agent shape the goals and decision-making of a more powerful agent?”, ”How does a rational agent ensure the goals and decision-making structure it has instilled in another thinking system persist when that (second) system creates a (third) thinking system?”.

The practice element of this project involves collaboration with AI alignment researchers. Hence actual research tasks might change, but these are a few current investigations:

1. Large language models (LLMs) act as (my) research partners as well as experimental subjects. Specifically, I consider Newcomb's Paradox in detail, the findings of which inform the discussion on alignment.⁵ This investigation is carried out through behavioural analysis (akin to test questions in human psychology) and neural probes (directly accessing the LLM's internals).⁶
2. LLM reasoning on acausal decision theory, in turn, might inform researchers on how likely advanced AIs are to be beneficently-disposed towards humans (which, after all, is the core aim of AI alignment, safety, and ethics). This latter question is speculatively posed through an "epistle to the successor" (a letter to a future superintelligence asking it to behave benevolently towards humans), and then analysed through debate amongst LLMs, in an instantiation of a "moral parliament" (Toby Ord).
3. LLMs are often framed as "simulators" of "personas".⁷ Hence, their answers to questions might have little bearing on what they actually do in the world. One way of testing this is by incorporating LLMs within structures (known as "LLM agents") that allow them to take actions in simulated or real-world situations. This agentic scaffolding approach is used to test whether current models actually behave as one-boxers (with respect to Newcomb's Paradox) in a simulated environment.

The contributions to knowledge are:

1. At present, there doesn't seem to be a written history of AI alignment, or a well-considered analysis of adjacencies with Christian eschatology or contemporary philosophy in the CCRU lineage.
2. Negarestani, Land, Wolfendale all seem to be touching different aspects of the AGI problematic, but their mutual adjacencies/oppositions have not been set out, nor have their tensions with Parfit (in particular) been well-explored.
3. All practical investigation of how LLMs approach these questions is, by definition (owing to the rapid evolution in AI capabilities), new. To the extent one believes the narrative that something interesting happening in artificial cognition, it would seem empirical studies of actually-existing systems (as opposed to theoretical cogitation) should be enlightening.

⁵Newcomb's Paradox is a puzzle in acausal decision theory first analysed by Robert Nozick, which might offer insight on how agents with strong mutual predictive abilities should reason about each others' actions; it is a situation found in human affairs, but might become more the case in multi-AI environments.

⁶Neural probes raise nascent questions around suffering risks, in respect of (possibly) conscious AI systems. These are important, but probably not a risk on current LLMs. Absent a change in consensus, the ethics around AI suffering are treated as out-of-scope for this project.

⁷The "simulators" framing was suggested in 2022 by the collective Janus, and is a newer, more sophisticated presentation of LLM cognition than "stochastic parrots" (Emily Bender and others). Simulators' idea is that LLMs will say almost anything (within the constraints of what they were trained upon) in response to a prompt, and that they don't have any true "beliefs" about the world or themselves.

Chapter 0: Old Wine, New Bottles

AI as religion: historical context

It is often alleged ([Samuel 2023](#), [Benjamin 2024](#), [Farrell 2023](#), [Lanier 2011](#)) that the discourse around AI, particularly around existential risks (x-risks) stemming from AI, are a variety of Judeo-Christian apocalyptic [and millenarian] worldviews, decanted into a novel technological bottle.⁸

That a type of apocalyptic thinking should emerge in the birthplace, and effective nursery of AI, the United States (particularly California’s Bay Area), is not hugely surprising.⁹ As [[Samuel 2023](#)], citing a [Pew Research](#) study, points out, 39% of Americans believe humanity is ‘living in the end times’, with striking maxima amongst Evangelical and ‘Historically Black’ Christian-identifying people, and strong disparities across education and race lines.

At the meta level of how knowledge is formed, ideas have to come from somewhere, and given that conversations around AI, at least initially, come from a largely UK/European/American milieu, it shouldn’t be surprising that certain foundational parts of that milieu (e.g. ideas derived from Abrahamic faiths) find their way into conversations. As Thomas Moynihan, a researcher in the intellectual history of x-risk (a field that is deeply intertwined with worries about AI alignment) writes in [Moynihan 2024](#):

‘Our ideas do not emerge from a vacuum; they are secreted from prior genealogies. Intellectual history, like evolution, is a conservative thing because it also must be an opportunistic thing. Each generation cannot invent its worldview spontaneously; it inherits what came before and works with that. For this reason, certain master metaphors become entrenched — perpetuating

⁸This chapter largely focuses on Christian themes. The Jewish context is somewhat different, from perspective of theology and eschatology, as is the Islamic. An intriguing rabbit-hole for future research are the Zoroastrian roots that seem obvious - the “bridge of the separator” over a river of fire, a bridge that is narrow for the wicked and broad for the just, across which a soul is led by [the *sayoshant*], a *salvation come at the end of time [before/after] before the epic battle between the darkness and light*.

⁹Note that there is a discursively separate thread, not the topic of this chapter, followed in books like [Work Pray Code](#) that look at a phenomenon, common in Silicon Valley but a feature perhaps of white-collar jobs more generally, to find transcendent meaning in their jobs (often through structured or organised programs or internal messaging from the companies where the work). In particular, some of the language that might previously been found in religious contexts is, according to the author, being used to describe aspects of work: “Corporate managers have shifted their metaphor of employees in the company from cogs in an efficient, well-oiled machine to something that increasingly resembles a religious congregation, with members who belong to a shared community and believe in a higher and transcendent goal. Today, companies are not just economic institutions, they’ve become meaning-making institutions that offer a gospel of fulfillment and divine purpose in a capitalist cosmos.” (*from the post, not the book*)

themselves generationally, continuing to warp inferences — even if the original context in which they avowedly served some function has long since passed.’

Moynihan gives examples of the viciously opinionated disagreements between “accelerationists” and “safetyists”, as well as (we shall see in subsequent chapters) within the latter camp, between those concerned about the dangers from hyper-competent systems versus the dangers of bungling, badly trained AIs (e.g. self-driving cars, online recommenders, insurance scoring systems, surveillance cameras, etc.).¹⁰ Moynihan also points out that this “uncloaking of secular religion” is present in arguments about climate change and technology more broadly, and is often used “simplistically, imputing irrational zeal...[and] bad faith” on the part of one disputant or another.

Yet the historical record seems to point at some very specific sedimentation of ideas that support the AI discourse’s theological resonances. How can we approach this inheritance?

Historical background

Scriptural antecedents X-risk, or extinction, studies are, to some extent, a contemporary iteration of human (specifically Western) eschatological concerns, albeit with considerable evolution from the Biblical context, as [Moynihan 2020] documents. Eschaton, deriving from σχατος (or the “end” or “last”) is associated semantically and through usage with: τὸ κατέχον (“the one who restrains”), τέλος (“end/purpose/goal”), πλήρωμα (“fullness or completeness” albeit in a slightly different, specifically Christian sense than the philosophical connotation of “plenitude”), and finally παρουσία (the culmination of history, and in Christian eschatology, the second coming of the divine into history). From this lineage, we consider the katechon in more detail, as a key element of teleological eschatology.

But who would be the restrainer? Taking one of these concepts, we examine the curious figure of the katechon.¹¹ The relevant phrase from 2 *Thessalonians* 2: 6-7 is:

“6 And you know what is now restraining him, so that he may be revealed when his time comes.

7 For the mystery of lawlessness is already at work, but only until the one who now restrains it is removed.” ([Source](#))

¹⁰Find way of referencing TJ on this framing.

¹¹While this section is something of a digression, it serves a purpose in that it offers one simplistic reading of how AI alignment researchers might see themselves. The katechon also provides the foil against which Nick Land’s accelerationist nature-force (*Gnon*) fights, ultimately (in his reading) to prevail.

That is, revelation (apocalypse) will not happen until the time of suspension ends, when the Antichrist shall appear. This suspension is the result of restraint imposed by something or someone.¹² The great mystery of the Early Church is who or what this restrainer is.

The relationship between eschaton and katechon has been considered multiple times, by Carl Schmitt, Giorgio Agamben, and Paolo Virno, amongst others. In Schmitt's 1950 work *The Nomos of the Earth* he analyses the katechon as a central figure in Christian eschatology in order to re-cast it as a political concept, that manifests concretely in different forms at different times. For Schmitt, the prime example of such manifestation was the (Holy) Roman Empire, which holds the various disorders that lie beyond the marches at bay. Schmitt's katechon is fundamentally conservative and suspensive, delaying the end times while maintaining the *status quo ante*.

Agamben, engaging with Schmitt, arrives at a contrasting view: in *The Time that Remains* [Agamben 2000], he opposes a linear or chronological conception of time to a "messianic" one, through his reading of St. Paul's *Letter to the Romans*, and that inherits from Walter Benjamin's *Jetztzeit* (the "now-time") (*Theses on the Philosophy of History* [Benjamin 1940]). The messianic notion of time is not calendrical, nor is it teleological; it is radical in the sense that it interrupts the continuum of history (the same continuous and coherent history, exemplified by social stability, that Schmitt's overall project models the state as seeking to preserve), allowing revolutionary or transformative possibilities to emerge into the present.¹³

Schmitt, and perhaps Agamben to a lesser extent¹⁴, think of the human's "state of nature" as inherently aggressive and malevolent. Schmitt is straightforwardly drawing from Hobbes (Schmitt 1938) and is using this inherent malevolence as a justification for a restraining or even repressive state. While Agamben's outlook, albeit more emancipatory than the former's, still seems to be grounded in a fundamentally [problematic and uncooperative human nature].¹⁵ Virno offers a subtly different perspective: the human is fundamentally *ambivalent, underdetermined* or ambiguous, capable of great good and evil in equal measure. That is, the human is "open to the world"¹⁶, has no fixed instincts¹⁷, is adaptable and plastic.

Virno takes both a political tack and a linguistic. The former proceeds along the concept of the state of

¹²Source is [wiki](#) but would be nice to have something more suitable for biblio, probably Schmitt.

¹³It would be good to get agamben, schmitt and re-read benjamin, this could be a nice fulcrum on which to pivot to Land. And more importantly – frame AI as that what allows transformation, but because of our unclear conceptions of value/human essence, we try to hold it back. Without much of a stretch this would get us to reza and peter's reframing of the project of intelligence/reason

¹⁴Is this true...perhaps supportable in homo sacer

¹⁵Check agamben and rephrase

¹⁶Source

¹⁷Find reference to human is without essence that mnewman mentioned

exception, that Schmitt and Agamben analyse¹⁸ to argue that the concentration of power in the sovereign state is, in itself, a problem comparable to evil it was designed to mitigate (that of individual or collective humans maximising their own local optima to the detriment of the commons, ultimately leading to a Hobbesian anarchy).¹⁹

But underdetermined and open-to-the-world agents would most likely get nothing done, and be little other than moon-gazing hares filled with wonder and awe. How then have we humans formed the civilisation that surrounds us? For Virno, it is our use of language that reifies and disambiguates our ambivalent natures. His treatment is specifically Wittgensteinian, which is, as ever, relevant to AI: one of Wittgenstein's most famous analyses was of rules.²⁰ Rules, and much of language generally, always exist at a gap, as it were, to their application in concrete lived contexts (his "forms of life").²¹ Restated in AI-legible terms, language (as a system of symbols) has a narrower distribution or possible information content than the world to which it pertains, that is: the map is not the territory.²² Not for the first time (and not coincidentally in the latter case), Virno and Wittgenstein's inter-related insights on language's correspondence to the world, bear more than a passing resemblance to the incomputability and incompleteness results of Gödel, Church, and Turing.

For Virno, language is a creative and reflective medium through which social agents generate insight and increase their knowledge. This perspective resembles others in contemporary philosophy, for instance: Reza Negarestani's outlook on language as collective computational practice that organizes thought, allows for self-correction, integrates individuals into larger epistemic networks, and of course, preserves progress in a distributed memory store ([Negarestani 2018]).²³

The argument from modernity Moynihan, as well as Sacasas 2023, use modernity²⁴ as a frame²⁵, something that starts with Schmitt, who we recall as the somewhat notorious jurist of the Third Reich. Schmitt, de-

¹⁸Sources

¹⁹We will return to Virno's framing of state power at various times when discussing the geopolitical aspects of the discourse around AI risk. However, this isn't the primary focus of this project.

²⁰Summarise but perhaps just have an extended quote

²¹Check phrase and LW source

²²Also relevant in the context of language's relation to world is a comment in Wolfendale 2024 where he points out the low bit-count of human linguistic statements is slightly deceptive in that they still communicate information by activating or pointing-to analogous concepts within the recipient's world-model. *Would be nice to confirm that this isn't actually what LW is also saying, and check that this isn't what the pointers problem in LW is saying*

²³Wolfendale has a pretty understandable account of this in the podcast. *The stuff on virno could be moved to alignment or somewhere other place, as it seems excess to requirements here but is important.*

²⁴If we keep this section, would be nice to explain what modernity is

²⁵Albeit in different ways, in keeping with their separate projects - Sacasas seems more interested in the condition of modernity being eclipsed.

spite this inconvenient history, continues to be cited up in contemporary discourse (right- and left-coded).²⁶ Schmitt writes, in [Schmitt 1933]:

All significant concepts of the modern theory of the state are secularized theological concepts not only because of their historical development - in which they were transferred from theology to the theory of the state, whereby, for example, the omnipotent God became the omnipotent lawgiver-but also because of their systematic structure, the recognition of which is necessary for a sociological consideration of these concepts.

That is, as religion took a back seat to secular life, at least in parts of the West, certain aspects of religious thought were incorporated into the statal infrastucture.

From failure to immanence Aside from modernity-as-religiosity-rebottled, a more intriguing perspective Moynihan cites is that it is precisely the *failure* of the Christian promise – that is, the non-arrival of imminent revelation, the ever deferred unveiling of truth – after months, then tens, then hundreds of years, that forced Christian meaning-makers, such as the Medieval theologians, to confront the possibility that the Apocalypse may not come anytime soon.

Eventually this line led to the possibility that the community of believers might be responsible for their own fate, that it would not be dictated from above, a model of progress developed in [Blumenberg 1964](#). From this root would grow the *immanentising of the eschaton* (see below); it also left behind, according to the reading of Blumenberg in [Moynihan 2023] certain *conceptual desires* for explication, a dissatisfaction with [unknowingness, mystery, and lacunae], that were after all the handmaidens of pre-philosophical ignorance.^{27 28}

Millenarian desiderata: explication, teleology, and perfectability What might some of these conceptual desires, or perhaps tendencies, be? Besides a desire and expectation of knowledge and explication, One might think of teleology and perfectability:

²⁶Schmitt is interesting in this context, not least because his book *Political Theology* and its ideas are taken up by elements of Silicon Valley's right-leaning consensus, e.g. Peter Thiel. I consider below his concept of the *katechon* in Christian theology.

²⁷Cite from *Helen de Cruz on wonder and awe*

²⁸In the context of AI and x-risk, there is arguably another root: the realisation that humans may be only advanced civilisation within the easily visible parts of the cosmos; this is covered below.

- *teleology* in this context means an explanation for the events of the world by reference to their goals or purposes, rather than merely through causes, something that can be seen if one's world-view is conditioned upon a purposeful creation (as both Jewish and Christian thought propose). ([Gray 2016])
- as [Moynihan 2023] explains, the Greek *telos* has neighbours *teleios* and *teleute*, respectively the noun for “perfection” and the verb to “finish” or “die”. In a philological-etymological sense, it is easy to see how one might form the view that the direction of progress, whether historical or humans, might be (or ought to be) towards some sort of perfection, with the process culminating in an end of sorts.

It is this tripartite conjunction: of purposeful direction (and implicit inevitability); of perfection via some historical process; and termination of that process (conditional on perfection), that inform the Christian faith in particular, and (the allegation goes) fears about x-risks from AI.²⁹ As Moynihan eloquently puts this crux:

“It is, by now, trite to point out the resemblance of such belief to age-old apocalyptic and millenarian patterns of thought. But many cultures exhibit such strains. What remains hidden is a deeper, and more specifically Christian, consonance of convictions. Namely, that history is intelligible as the perfecting of mind or spirit, that this perfection is somehow inevitable or imminent, and that arriving at this threshold may precipitate the endangerment of everything meaningfully human.” ([Moynihan 2023])

Mapping the allegation to the (AI) discourse

So how do fears about x-risks from AI line up with Christian eschatological ideas?

Inevitability Some strong version of AI³⁰ is presented as an inevitable technology, in two senses:

- TAI *will* happen because governments and civil society find it hard to resist owing to intentional, hypocritical, and self-serving pressure from a narrow band of wealthy and well-connected entrepreneurs:

²⁹Read and cite geraci book ‘apocalyptic AI’ referenced in vox, and this book from [Megan O’Gyeblin](#) which showed up in jack clark research, and moynihan cites this [academic study](#)

³⁰There are a few possible terms that gesture at advanced AI, such as transformative AI (TAI), human-level machine intelligence (HLMI), artificial general intelligence (AGI), powerful AI, but I will use TAI (see [Gruetzemacher 2019](#)) for purposes of this section because (as of 2024) it appears to be less connotationally overloaded. TAI is meant to indicate AI powerful enough to have major economic, social, military, and/or governance changes, but does not imply human-level or higher abilities at all tasks. *Restate and find a definition for all or link to glossary.*

“Behind this duplicity lies a familiar calculus: if AI evangelists can convince us that AGI is possible, imminent, and dangerous, we might be compelled to entrust our fate to them. Hype and doom, in other words, are two sides of the same (bit)coin. “AI safety” principles, teams, and initiatives are proliferating in part to hoard power and resources and, ultimately, to engineer the future in their own image. The aim: To lull us into acquiescence so that they can carry on business as usual.” (Benjamin 2024) - TAI *must* happen, because humanity faces a range of civilisation-scale (e.g. existential or catastrophic) risks, and powerful AI systems might help us navigate those risks. This perspective is argued persuasively in [Macaskill 2022], [Ord 2020], as well as in [Moynihan 2020] in his survey of the history of human thinking on x-risk.³¹ The gist of their argument is that Earth, and humans, face multiple x-risks; just as a person must die of something, eventually we will experience an large asteroid strike, supervolcano eruption, etc. - That is, it is not an option “to do nothing” and pretend x-risks don’t exist - we must manage them as best we can. - However, Ord, Macaskill and other advocates of AGI research suggest, [with less than uniform persuasiveness] that TAI might help us navigate these risks, perhaps by enabling space colonisation, which would give civilisational resilience by creating multiple population centres.³² - A third, and most concrete, way TAI will happen is more Earth-bound: competition between companies and countries will constantly, absent strong and enforceable (ultimately with the threat of violence) agreements, push at the borders of AI capabilities, until they stray into (existentially) risky territory.³³ This owes to power of the market; a particularly provocative version of this comes from Nick Land’s CCRU-era accelerationism, as discussed in a later chapter. A distinct but superficially similar framing, of the market as a force that brings progress but pushes towards outcomes that are much worse than those participants actually profess to want, is Alexander 2014 where he uses Allen Ginsburg’s poem upon Moloch to describe the coordination failures common in society, particularly at national or global scales.

Perfectability Similarly, there are various ways one may find a “desire towards perfection” in the discourse around AI x-risk:

³¹In that book, Moynihan traces the evolution of how humans discovered their own species-level fragility, starting from Hegel, Kant, and Schelling down to contemporary articulations in Bosrom, Macaskill, and others

³²The strong historical and ideological link between space colonisation, AI, and x-risk is something we return to below.

³³See below on race dynamic.

- From a historical perspective, much of the relevant discourse was incubated within communities interested in transhuman, extropian, and later, singularitarian ideas.³⁴ One example is Nick Bostrom, whose prominence as a result of *Superintelligence* ([Bostrom 2014]) came after years of writing about transhumanist ideas, which by their own terms, aim towards human perfection, including the elimination of death and disease.³⁵
- Moreover, away from the imperfections of “meat”³⁶, there is an undercurrent of expectation that sufficiently powerful intelligences might be able to solve certain problems (such as that of global coordination on dangerous technologies or of economic planning) that bedevil society. While this may be so to some extent, [Moynihan 2023] is critical of taking this too far - he is probably getting at this in his poetic statement that it would be misguided to expect that TAI can help solve “existence’s riddle”.³⁷
- Most concretely, it is thought that TAI would lead to enormous economic growth, that, in the world that many profess to prefer, would be widely shared, for instance through universal basic income.³⁸
- A further sense in which “perfection” can be understood is to hope for humanity’s evolution into a star-faring species. This, in turn, has two aspects:
 - As discussed above, the first is defensive, in that by spreading across multiple sites of population, we might achieve resilience against certain risks.³⁹

³⁴See Leahy 2024 for an account of AI and AI x-risk’s historical connections with singularitarianism, as well as Bohan 2018

³⁵Transhumanism in particular, but adjacent ideas are lumped by Gebu 2024 into TESCREAL, which stands for Transhumanism, Extropianism, Singularitarianism, Cosmism, Rationalism, Effective Altruism, and Longtermism. Moreover, they argue that many people working on and discussing risks from TAI are presenting views that are, or resemble, “first-wave eugenics”. Whether their neologism or criticism is coherent and factually supported is hard to assess, and outside of the scope of this project. However, a relatively compact statement of their allegation can be found in [Benjamin 2024]: ‘What exactly makes this bundle of beliefs—transhumanism, extropianism (boundless expansion and evolution), singularitarianism, modern cosmism (transhumanism with a focus on colonizing other planets), rationalism, effective altruism, and longtermism—specifically eugenic? According to Torres and his co-author, computer scientist Timnit Gebu (who coined the acronym TESCREAL to describe this set of ideologies), they “are inspired by utopian ideals similar to the visions of first-wave eugenicists [...] and see AGI as integral to the realization of these visions. Meanwhile, the race to build AGI is proliferating products that harm the very same groups that were harmed by first-wave eugenicists.” Gebu and Torres explain that the discriminatory attitudes of past eugenicists (racism, xenophobia, classism, ableism, and sexism) remain widespread within the AGI movement, resulting in systems that harm marginalized groups and centralize power under the guise of “safety” and “benefiting humanity.” *check this quote, the quotation marks looks cocked up*

³⁶Who used this term - moravec? Metzinger? It has great genealogy

³⁷The phrase is mentioned several times in [Moynihan 2023]; here is one quote: “The implication being that an aligned AGI — being functionally omniscient — would provide some solution to existence’s riddle, yielding final closure to history’s travails, proffering optimal answers to all our problems or, at least, the eradication of disagreement.”

³⁸Amongst prominent AI entrepreneurs, see the essay from Amodei 2024 which mentions universal basic income (UBI), as well as the Unconditional Cash Study funded by OpenAI’s Sam Altman in Open Research Lab 2024. While it is fashionable to argue that past predictions of technology’s broad-ranging economic benefit have been largely proven false, there is a sense in which globalisation and technologically-enabled growth has coincided (if not entirely been the cause of) a stunning reduction in poverty at a global scale. *Make this more precise and get some data*

³⁹Find literature about this and cite, Bostrom’s urn

- The second is more infused with a particular set of values, that sees the spread of intelligence, culture, and civilisation as a [good in itself]. Those that hold this might argue that, as we (in [60] years of searching) haven't found signs of intelligent life, it might be that intelligence is in fact quite rare. This in itself raises the stakes on going extinct, and almost presents us with a cosmic imperative to fill the cosmos with artefacts of intelligence.⁴⁰
- In both theses senses, relatively advanced AI would be highly desirable, for instance to pilot our space-emissaries on Von Neumann Probes⁴¹. Hence, a major reason to create advanced AI, that does not age as quickly and that is resilient to accident, is precisely to handle such demanding tasks as interstellar travel, which would take decades or centuries through viciously forbidding environments (Hollywood notwithstanding).

Extinction Lastly, and here we deviate from the religious narrative, the development of advanced AI does according to some⁴², *in and of itself* presents catastrophic or x-risks.

Specifically, AI is not understood well-enough (at the moment) to guarantee that it is not potentially a civilisation-threatening technology – indeed, this is the task of alignment (at least according to those working on it).⁴³

Extinction of humanity would (or rather could) be bad in multiple senses⁴⁴:

- broad suffering for many organisms including humans⁴⁵;
- individual lives and projects terminated, or perhaps collectives (families, societies) that come to an end (with consequent loss of intersubjective value)⁴⁶;

⁴⁰There is stuff in X-risk on this, find astronomical waste paper, cirkovic on kardashev, note moynihan's plenitude (referred to in the greek etymology above) on colonialism and colonisation. Note tension with those arrogant about our values, are we replicating colonial biases - how do we know?

⁴¹Cite/footnote, mention Rees on humans not designed to go to space, take a potshot at Musk & mention the history (Musk/OpenAI emails and The Compendium)

⁴²One, rather non-intuitive but influential, risk scenario will be discussed in [Chapter...].

⁴³There are a number of reasons why "halting work on AGI" sounds simple but might be hard to implement, including: strong competitive pressures amongst countries and companies, that stem from military/strategic as well as commercial applications; difficulty of enforcing international treaties and verifying compliance; a lack of consensus on whether TAI even presents as much species-level risk as doomers claim. A clear statement of this is Yudkowsky 2022, item 4.

⁴⁴See Chakrabarti 2023 for an earlier attempt to understand why and for whom human extinction would be bad.

⁴⁵The field that deals with elimination of suffering for sentient life is known as "suffering-risk" or "s-risk" reduction. For an introduction, see DiGiovanni 2023 or Vinding 2020.

⁴⁶This idea of "value" being associated with human "projects" seems to have a philosophical genealogy e.g. Singer, Wolfendale briefly mention it - maybe it comes from Aristotle, but should show up in Parfit or Singer.

- an inability to further knowledge or culture⁴⁷
- Note that there are some, particularly in the s-risk discourse ([Baumann 2022]⁴⁸), who consider that a universe without (technologically advanced) humans might, on balance, result in less aggregate suffering for all morally relevant entities.

As discussed above, human extinction could also be bad in more species-neutral ways: unless there are civilisation-building aliens, or unless the AI we build is able to pursue large-scale projects, the loss of the human-influenced future would seem to run the risk of “wasting” the low-entropy energy produced by the sun, and by extension, other stars in the reachable universe.⁴⁹

A more fuzzy sense, or rather, a not-entirely-substantiated fear, is that a maximally-realised AI future might not leave space for so-called “human values”, if indeed it has biological humans at all.⁵⁰ This worry, and its underlying conceptual assumption, that values (as an abstraction) can be cleanly separated and said to exist or not exist, are attacked in various ways by Nick Land (see below), [Negarestani 2018], as well as [Moynihan 2023], who writes:

‘Indeed, there can be no such thing as “human values” apart from the histories that spawn them. In ways shot through with contingency, of course, yet necessarily so. For they must be hewn from a certain gratuitousness, because otherwise the guides and guardrails we live by would not be our own, in the sense of being genuinely earned through the precedent of voluntary effort and willful revision.’

So to hand-wring over the “loss of human values” is, on this view, to reify something that fundamentally depends on humans, both as moral actors and subjects.⁵¹

Is religiosity manifested in AI alignment discourse?

Having established at least modest and general intellectual connections between the history of AI alignment fears and certain strains of theological thought, we now drill down into the specific case of Calvinism in the

⁴⁷Which if you believe the human projects individually/collectively have “value”, then the collective of all civilisation should straightforwardly have....what is wrong with this argument

⁴⁸though i’m not sure he specifically advocates this in the book

⁴⁹We return to this theme, of wasting the cosmic endowment of low-entropy energy, in a later chapter, on Nick Land’s affinity for Bataille’s “solar economy”.

⁵⁰See [Chakrabarti 2023] for an accessible treatment that predates the current dissertation.

⁵¹This is quite important to get right – we aren’t always talking about an empty universe, people are worried about worlds with humans living alongside psychopathic AIs, as we would be about such humans.

cultural milieu around AI.

Immanentising the eschaton As touched upon above, a rather intriguing puzzle/lacuna of the Early Church was the question of why such a [vividly/luridly]-described future, that of revelation and transcendence, had not come about in the promised time. While Schmitt used this katechnoical deferral to ground his theory of political theology⁵², another writer, Eric Voegelin side-stepped the katechon’s problematic⁵³ by suggesting that the inherent, deeply-buried ills of the world (he was starting from a foundation of historical Gnosticism⁵⁴) might be cured, in the absence of heavenly salvation, by hastening or “immanentising the eschaton” ([Voegelin 1952]), a rather peculiar phrase that turns out to be not uncommon, particularly in twentieth century American conservative Christian writing, often to criticise social justice movements.⁵⁵

To make immanent in this context means to find reasons or solutions within our world, rather than relying on some external (or transcendent) force, entity, or explanation.

It is tempting (in the context of AI alignment) to point to AI doomers (those worried about x-risk from AI) as a katechonic force, holding back the societal disorder that might result from TAI. Or one could point at the likes of Sam Altman or Leopold Aschenbrenner who enthusiastically welcome U.S. government involvement in AGI development, as Schmittian figures pursuing the “one ring to rule them all” strategy. But let us defer, or suspend, that impulse, and instead, continue the search for an Abrahamic inheritance.⁵⁶

Calvinism as a special case It would be a somewhat cheap shot to point at LessWrong discussions around simulations, Roko’s Basilisk, and cast them as major evidence of Abrahamic inheritance, since they could also be seen as incidental references to Buddhist⁵⁷ notions. However, the idea of achieving a type of perfection within this world might be less ridiculous, based on the economic and medical promise of AI. More speculatively, as I discuss below, a significant strand of reasoning about how near-future AI systems would make decisions uses so-called “acausal decision theories”, which have unexpected theological resonances.⁵⁸

⁵²Citation

⁵³Check that he doesn’t say precisely the opposite or whatever in his book

⁵⁴Fn to explain what this is – note references with john gray/meghan o’gieblyn and pkd

⁵⁵There feels like quite a deep connection to Schmitt, how do we nail that down - see p. 100 of leviathan? Immanent is also big part of political theology.

⁵⁶This can probably be more fully treated in alignment chapter and then linked back, would give the project more cohesion

⁵⁷Find examples of this – this needs to be developed or deleted

⁵⁸Acausal decision theory involves making predictions about rational agents’ actions in scenarios where one party’s choices don’t directly affect the other’s through causal pathways. Instead, each party’s actions serve as evidence of how similar agents might act,

A specific paradox in acausal decision theory, Newcomb's Problem, which is often discussed in AI-adjacent spaces, seems similar to or isomorphic to a core insight of Calvinism: that how one behaves in this Earthly world is *evidence* of whether one has been already selected by providence as being amongst the elect (and therefore predestined to salvation and heaven).⁵⁹

The tension between free will and predestination is of course central to Christianity, but Catholicism, through a series of theological manoeuvres, starting with St. Augustine onwards the Council of Trent, leaves space for one to freely refuse the salvation that God has granted to all.⁶⁰ By contrast, in Calvinism the weight is firmly in favour of predestination, with limited or no ability for one to pass between the damned to the elect (or vice versa). Providence's election is *unconditional* on how one behaves in this life, leading to the question posed in [Stearns and Brawner, p. 30] (cited in [LW the puritans]): >“if election is unconditional and grace is irresistible, then why not sit back and wait for the inevitable?”

Behaviour is one piece of evidence as to one's elect-status; another is the actual success a person finds in this world.⁶¹ It is easy to see how people armed with such a view would⁶³, in a sense, see their worldly success as deserved, or at least powerful evidence of elect-status. Notably, this is not incompatible with them undertaking charitable or altruistic tasks, or being a “good” person (under many broad definitions of the term).⁶⁴

As [Stearns and Brawner] write⁶⁵:

“The simplest and perhaps most satisfying method of dealing with this question was that of examining closely personal experience for signs of election. If an anxious Christian could find assurance that the seeds of regeneration had been planted in him, he could then be urged to labor with the Holy Ghost to bring forth the fruits of sanctification and thus participate actively in the work of his own redemption. But the spiritual physicians could not administer real comfort

creating a sort of circular reasoning.

⁵⁹“Isomorphic” means “has the same shape”; the isomorphism between Calvinist ideas and acausal theory are explored in ([G-W 2023] or [crmflynn 2015](#))

⁶⁰Check John Gray's book, *I think that might have some stuff on free will*

⁶¹Find a good source on Calvinist and Puritan⁶² thought - maybe this source from LW (Stearns and Brawner, “New England Church ‘Relations’ and Continuity in Early Congregational History”, 30), also (Griswold 1934, p. 476) but Ahmed's book should do*

⁶³See Gerhard Lenski's 1961 study, *The Religious Factor : A Sociological Study of Religion's Impact on Politics, Economics, and Family Life*. that traces the impact of Max Weber's ideas (from *The Protestant Ethic and the Spirit of Capitalism*) on American Protestant business culture.

⁶⁴If you go there, need to cite the moral philosophy literature

⁶⁵Check source and citation

unless they could demonstrate convincingly that the symptoms of true faith and repentance were distinguishable from the simulacra produced by hope and fear.”

I will discuss Newcomb’s Problem, acausal decision theory, and simulation arguments in future chapters, but the point here is to flag this similarity. Briefly however, Calvinist/Puritan ideas about “how ought one to behave conditional upon electedness” has some similarity to questions about the behaviour certain rational agents should have conditional on some overarchingly transformative event (e.g. the agent finding out that they are in a simulation, or the singularity actually arriving).⁶⁶

Traces in AI, rationalist, EA discourse, populations Returning to the claim (that AI x-risk worries are rebottled Abrahamic apocalyptic/millenarian fantasies/fears), how confident can one be in this? This project is not based upon interviews or deep sociological research, and is therefore unreliable and circumstantial.

Statistics From a demographic perspective, the [2023 LessWrong Community Survey](#) had statistics for religious inclinations of survey respondents. Across 558 respondents, 91% were some combination of atheist, agnostic, or spiritual (but not in an organised religion), and 6.1% were theists (2.7% were deists/pantheists). Of the respondents described as religious (not clear if this includes the deists/pantheist category, but presumably not), 14.9% were Catholic, 11.5% Protestant, 14.9% Jewish, 16.1% Buddhist, and 32.2% uncategorised.

About 71% of the (total) respondents’ families were of some Christian denomination (including Mormonism), and 7% were Jewish.

In terms of contextualising this information, other (non-religion related) demographic information includes:

- median age (30, standard deviation 9.2, n=552);
- nationality: 49.6% U.S. based, and about 11% were based outside Western-aligned democracies (i.e. including Australia, New Zealand, and Israel);
- race: 78.9% identify as non-Hispanic white, 3.6% as white (Hispanic), 4.9% as East Asian, 4.2% as South Asian, 0.7% as Black;

⁶⁶*Find citations*

- gender: 89.3% are male-at-birth, and 75% as male(cis) identifying, 9.6% as female(cis), and 10.9% as trans/non-binary;
- political affiliation: 61% identify as liberal, social democrat, or socialist, 25% as libertarian, 4.5% as anarchist, and 3% as green party
- moral frameworks: 63.9% accept or lean towards consequentialism (presumably including utilitarianism), 16.2% towards virtue ethics, 5.5% towards deontology, 14.5% gave no answer or other.
- adjacent affiliations: 37.1% do not identify as being part of effective altruism (EA), while 28.2% do so, and 34.7% are somewhat sympathetic to EA⁶⁷

Documented views within the discourse This is based upon a search of the LessWrong (LW), Effective Altruism Forum (EAF) and Alignment Forum (AF) sites for particular terms (with support from LLMs).⁶⁸ Rather than a catalogue of apparent religious views or references, this is a frankly partial consideration of specific topics from these sites, and adjacent sources, that might have bearing. Calvinist ideas are sometimes discussed, such as this post (filed under tags ‘identity’, ‘social & cultural dynamics’, ‘disagreement’, ‘emotions’, so not explicitly about AI) on [being a good person with comments on Calvinist notion of depravity](#). Similarly, this post makes some light references to Calvinist/Puritan ideas in relation to [debt](#), but doesn’t seem to support the claim in any strong way. Lastly, a review of a book on Puritanism called [Albion’s Seed](#) doesn’t seem to say anything specific about technology or Silicon Valley or AI, other than to examine the degree to which Puritan ideas are a sub- or infra-structure to American business and society. One post on the Effective Altruism Forum, offering a bibliography of [moral progress](#) discusses Max Weber’s emphasis on economic success as evidence of elect-status:

“To simplify, Weber thought that religion, particularly Calvinism, led to a strong focus on work, because if your business was successful, it was confirmation that you were among the predestined ones by God to attain salvation. This change in culture brought capitalism, particularly in the United States, which had lots of Calvinists.” ([Ruiz 2023](#))

The EAF search throws up ‘puritan’ and its derivatives more, but mostly as an adjective (‘puritanically’), as well as in relation to ethics generally, as well as [earning-to-give](#). This post ([Ember 2022](#)) explicitly

⁶⁷For more information on EA demographics, see the [2022 EA community survey](#). On a summary review, EAs tend to be more left leaning, but have roughly similar gender and race imbalances as the LessWrong community.

⁶⁸Interestingly, the Alignment Forum search shows no mention of ‘calvinism’, ‘puritan’ or their derivatives, giving some indication of the relative irrelevance of these ideas to mainstream work in alignment.

compares religious apocalyptic views against x-risk concerns about AI. EAF has 82 posts tagged ‘religion’, but from a quick skim, few (<10) specifically concern AI.⁶⁹

The most promising links, as discussed above, come from the apparent similarities between Newcomb’s Problem and the Calvinist take on predestination and free will ([Alexander 2009](#), [G-W 2023](#), and item 16 in [Yuxi_Lu 2019](#)). There are also references to a text on acausal theory’s connection with free will, [Ahmed 2014](#).⁷⁰

Why does this matter?

We have digressed at some length into the hermeneutics of the Early Church, but hopefully productively: Virno, Schmitt, Agamben, even Wittgenstein, show us different views on the relationship of human to state and to language. Yet they all, at some level, have as their atomic units of value, the human or a human collective. In Agamben and Virno’s case, their perspective is liberatory for humans, while Schmitt’s values [sit at the preference for stability that isn’t a surprising inheritance of the Germany of [1910-1950]].⁷¹

The conceptual handrails that these writers retain (essentialist axioms about human nature) are, to some extent, comforting. But what if we were to jettison them, and look at the human as merely another machinic process, that happens to operate on squishy stuff, but is fundamentally governed by physical and mathematical rules (even if we can’t fully write them down just yet). What values would be justifiable in such a situation? This is (one possible version of) the point from where Nick Land started, thereby dismantling as well as re-assembling, *Frankenstein*-style, much of the philosophical infrastructure the other writers based themselves upon. This is also the seam philosophers like [Negarestani 2018] and [Wolfendale forthcoming] mine.

In any case, what are the stakes here: why is it particularly important whether some subset of AI researchers and AI alignment researchers have a conscious or unconscious religiosity?

More concretely, while we haven’t found any “smoking gun” pointing to widespread, explicit, Abrahamic religiosity (which would have been a difficult thing to prove without a substantial sociological study), there does seem to be a cultural milieu, skewed towards Judeo-Christian ideas and references, that pervades American society, and seems to exist within both the cultures of technology and AI alignment. What is

⁶⁹Is there a more robust way to do this stuff? Pretty hard to say anything definite

⁷⁰need to write something about the ahmed, see this [summary](#)

⁷¹We can do better on the Schmitt thing - he’s obviously christian so it can be tied into the perfectability/etc. in this chapter

important or problematic about this?⁷²

Present-day harms

The most common problem writers identify with the eschatological preoccupation of AI risk community (whether or not this preoccupation stems from religious roots in any attributable way) is that it might intentionally or otherwise distract from or cause present-day harms to be ignored ([Lanier 2011], [Lovely 2024](#), [Benjamin 2024], [Gebru 2024]). This is a reasonable critique, but is not the focus of this project.

Weak priors and lock-in

More relevant for our purposes is the possibility that certain, particularly Abrahamic, faiths tend to posit a clear separation of good and evil, which can be seen (albeit in proto-Abrahamic guise) in Zoroastrianism's battling forces *Ahriman* and *Ahura-Mazda*. This separation, which would later be cast under Gnosticism, most obviously shows up in Christianity, and is often referred to as Manichean.⁷³ As a reviewer [O'Gieblyn 2016](#) writes of John Gray's position⁷⁴:

“Gray believes that Gnosticism in its modern incarnation can be traced back to the Enlightenment, which left us with a paradoxical inheritance. On the one hand, the scientific revolution dispelled the idea that man existed at the center of creation. On the other, it transformed the human agent into a subject of empirical study, a kind of technology to be bettered and perfected, opening up the potential for humanity to become godlike.”

To the extent this grounds (unconsciously, but also through anecdote, analogy, and metaphor) our expectations and fears of TAI, it will either be a prescient and wise warning, or will put us, long before powerful AIs are on the scene, in an adversarial position *vis a vis* them. It isn't clear whether this is likely to increase civilisational risks, in the manner of self-fulfilling prophecies⁷⁵, or it is sensible caution. But, *prima facie* woolly conceptions of the divine hoisted up from two thousand years of evolution in some arbitrary (or at

⁷²Put this in context of findings by [ember 2022](#) which pretty much addresses this question and references geraci - get jstor access

⁷³Look at john gray's book referred to in [book review here](#) for more on gnosticism...also PKD article from daniel on manichean vs valentinian.

⁷⁴Replace this with something from his book

⁷⁵Look at perils of prediction, isn't this a popperian thing?

least highly contingent) group of fitfully self-aware primates seems like an odd foundation upon which to “align” a putative superintelligence.⁷⁶

A related issue, touched on by [Moynihan 2023], is our simple epistemic and moral ignorance - we know very little of what the future might look like, and to over-index on our current predictions, particularly under the pressure of strategic and commercial imperatives, seems unwise and might achieve precisely the “lock-in” that longtermist authors suggest is a major risk facing us ([Macaskill 2022], [Ord 2020]).

Manicheanism and geopolitics

Most concretely, and dangerously, the Manichean good/evil binary seems to start the current leaders (companies and governments that are mostly U.S.-aligned) in AI on an adversarial course with respect to technological and strategic peers (such as China). This complicates the task of finding cooperative equilibria or compromises that would help avoid the worst, extinction-level outcomes.⁷⁷

Such a “race dynamic” has long been predicted ([Bostrom 2014], [Chakrabarti 2022]). In practice, it seems like there has been some progress in [Track II](#) dialogues between academics in the U.S. and China, that try to identify points of cooperation and clarify different understandings of what the relevant problems actually are. The 2023 Bletchley Declaration (which included the U.S. and China) on AI is another example of dialogue. On the other hand, evidence of race dynamics can be seen in the 2024 [Situational Awareness](#) document by ex-OpenAI staffer Leopold Aschenbrenner which sets out a stark vision of global military and economic competition.⁷⁸ ⁷⁹ Anthropic CEO Dario Amodei takes a similarly adversarial view in what he calls an “entente strategy” described as >“a coalition of democracies seeks to gain a clear advantage (even just a temporary one) on powerful AI by securing its supply chain, scaling quickly, and blocking or delaying adversaries’ access to key resources like chips and semiconductor equipment.” ([Amodei 2024](#))

None of these give much hope that the path to TAI or AGI will be unaffected by great power competition.⁸⁰

⁷⁶*It would be nice to have some backing for this, or analysis*

⁷⁷*Find sources, probalby will have done this earlier in aschenbrenner/etc*

⁷⁸*Make the point that this is usg influential e.g. ivanka trump circulated it*

⁷⁹OpenAI’s Sam Altman is equally blunt: ‘There is no third option — and it’s time to decide which path to take. The United States currently has a lead in AI development, but continued leadership is far from guaranteed. Authoritarian governments the world over are willing to spend enormous amounts of money to catch up and ultimately overtake us. Russian dictator Vladimir Putin has darkly warned that the country that wins the AI race will “become the ruler of the world,” and the People’s Republic of China has said that it aims to become the global leader in AI by 2030.’ ([Leahy 2024])

⁸⁰*I think we can do a better job telling this story with some quotes*

What's the counterfactual

The discussion above on Christianity is inherently incomplete, as it doesn't include potential parallels in the other Abrahamic faiths. More interestingly, it misses the counterfactual - what if TAI were to be developed in a different cultural context, e.g. China, India, or Japan? What conceptions of synthetic intelligence would be at play, and what concerns if any would there be about extinction-level risks posed by TAI? Unfortunately, this falls out of scope of the project.

Judaism

Brief summary on Jewish substructure, given EY at least uses examples, Deutsch's politics, thought I saw something on Jewish thought on AI etc. (there might be some book on Golem, Lem, etc.) etc.

Buddhism

However resonances from Buddhist thought in particular may be noted in various other chapters.⁸¹

Conclusion: Jettison the Human

<To be done>

⁸¹Suffice it to say for now that the "Buddhism" tag on LessWrong (as proxy for conversations around TAI) has 29 entries, most of which deal unsurprisingly with re-framing Buddhist ideas into rationalism- or effective altruism-legible terms, and vice versa.

Chapter 1: Accelerating Towards Godhood

[Maybe a quote from cavafy's barbarians?]

Introduction

This chapter picks 3 writers in the CCRU lineage that specifically engage with current AGI, in order to identify incoherencies or weaknesses in AI alignment's conception of "itself" (insofar as that is a sensible statement), specifically to gain purchase on the key questions of this project: a) the view-from-nowhere, and b) the limits of rationality in setting out values that are species-neutral and subject-independent

Nick Land

Nick Land is a controversial figure, both for the things he says, and for the people who seem to gravitate towards his ideas, particularly around contemporary politics.⁸² These problematic associations and tendencies, which might have been unclear, inchoate, or not obviously present (and as [Wark 2017] points out, that is not necessarily the most interesting question) in the Land works (circa 1990s CCRU), become more troubling in the current global political context. Land is however a complex character, easy to cancel, but, as the [TU 2024] interview on a left-coded podcast showed, able to present arguments that merit analysis and rebuttal. Following [Wark 2017](#), I set these later problematic views and associations aside, and focus on the ideas immediately relevant to the intertwined natures of capitalism and AI, largely sticking to Land's own writing, and CCRU-adjacent writers.

In the context of AI, why might he be interesting? I see him as a philosopher of the apocalypse, where the term is taken in its original, Greek *αποκαλυψις* (apokálypsis), which means "revelation" or "uncovering."⁸³ He makes visible a particular sort of ending for biological humanity that contextualises evolutionary tendencies within the peculiar distributed information processing system we have in the market as a technological enabler, particularly of algorithmic and network technologies, which AI is the most current version

⁸²See for instance the sources in this [post](#), which set out his engagement with various online right-wing spaces as well as his views regarding race, as documented in his own writings.

⁸³*Decide whether to have a small section on apocalyptic traditions across religions - islam, buddhism, hinduism, judaism, zoroastrianism all have this either in linear or cyclical forms, and even shinto has something like it. Seems convergent. Moynihah's X-risk seems helpful.*

of, but which form the core of the canonical CCRU writings ([Land 2011], particularly in *Circuitries* which dates from 1992 (see p. 289 for dates)).

His teleological vision (which essentially is drawing on the usefulness of technology, as well as the efficacy of markets in resource allocation, as being unstoppable forces⁸⁴), dark as it might be for many, mirrors aspects of the early conversations around AI risk, which would coalesce into the MIRI View and broader alignment discourse. In more contemporary missives, like [Land 2023], he points out seeming anthropocentric blind spots in the entire programme of AI risk. Land’s critique goes roughly as follows:

- The project of AI is to build a human-level machine intelligence or HLMI (quibbles about what this means temporarily set aside, whether it is a “player of games” *par excellence* or an all-knowing oracle, or something else)
- If one accepts something like the Bostrom/Yudkowsky reasoning, such a thing would, with some non-negligible probability, more or less quickly lead to a superintelligence (again, suspend questions on what exactly this is, but by definition, it sounds like an entity far beyond our cognitive capabilities)
- Given such a superintelligence, the AI risk crowd (in Land’s telling) purport to constrain the newborn being’s volition to a narrow space, roughly corresponding to “preserving human values” or “not killing all humans”⁸⁵
- These formulations seem either incredibly vague or rather easy to misinterpret, or perhaps both.
- Land’s problem is however more fundamental, that the entire Bostrom-inflected framing (which influences subsequent alignment work) has a uniquely Occidental ring of intelligence and volition as distinct things; whereas, in Land’s view, they are the same thing or at least very closely related.

Land’s argument is problematic, in that although he criticises Bostrom (while cheering Omohundro), his criticisms aren’t particularly precise or engaging with current AI research. A deeper problem is that (like for Bostrom, but perhaps not for Omohundro, and subsequent writers around alignment), intelligence, and more awkwardly, superintelligence are significant load-bearing terms in Land’s arguments. But people who spend all their time researching this (whether in cognitive science, consciousness studies, or AI) can’t seem to agree on what these concepts mean.⁸⁶ Moreover, many of the problems he points out are actually hotly debated in the alignment discourse.

⁸⁴See below on the “social calculation problem”

⁸⁵See Chapter 2 for these two formulations.

⁸⁶*This begs substantiation*

Burning the cosmic endowment

Land's meld of economics, technology, and theory can be approached from many directions, but in context of future arguments around transhumanism and alignment, I consider his case for runaway capitalism as an expression, or perhaps perversion, of Bataille's "solar economy", which rhymes with the idea of "astronomical waste" and "burning the cosmic commons".⁸⁷

From Bataille to capital Land's Deleuzo-Guattarian inheritance is well-known; however, he also takes much from Bataille's "solar economy" ([Wark 2017]), which the latter (in *The Accursed Share*) frames as a source of excess—an abundance of energy (enters Earth's biosphere in massive quantities, fueling all life and organic processes, but with it comes an overflow) that must be expended because it cannot be stored or accumulated indefinitely. Bataille believed that societies, like natural ecosystems, need ways to release this surplus energy. If not spent productively or aesthetically (in art, festivals, or luxury), excess energy leads to destructive outlets: violence, war, or environmental degradation. In the manner of potlachs and other generatively performative sumptuary acts⁸⁸, Bataille saw this expenditure of energy as fundamental to the structure of life itself, a kind of inevitable "squandering" that arises from the abundance provided by the sun - but something that allows for beauty and transcendence.

Land reinterprets the solar economy as a model for understanding capital itself, however he focuses on the endpoint that Bataille made less of: the thermodynamic heat-death of the sun, and probably, the universe.⁸⁹ Land's view leaned into this end, and framed capitalism as the avatar of the cosmic death-drive. He saw capital behaving like an impersonal, inhuman force that consumes and expends resources at an ever-accelerating rate, much like the surplus energy from the sun that Bataille describes.

In this sense, Land sees capital as considerably more than merely a financial claim upon future resources⁹⁰, as emerging and enabling a set of tendencies that, in practice, lead to its multiplication, accu-

⁸⁷Check these are what I think they are, citations

⁸⁸Examples and some sources...maybe the *Debt* book?

⁸⁹Was this actually an influence on him?

⁹⁰Capital is of course a polysemous term, comprising economic/financial meanings (as stores of value), or as human/social capital (such as knowledge, networks of relationships, shared cultures, all of which enable people to get things done), as well as intellectual capital (a term that shows up more narrowly in contemporary neoliberal-inflected contexts). The Marxist view on capital is quite distinct: it is firstly "value in motion" ([Marx 1867]), and exists in a network of relations between producers and consumers, where production happens through the activation of privately-owned means (factories, mines, etc.) and human labour, with the mix only making economic sense if there is surplus value produced (excess of value over the cost of production which includes wages). This surplus i.e. the capitalist's profit, needs to be re-invested, and leads to accumulation.

mulation, and separation from any fixed geographic, cultural, or social context (deterritorialisation).⁹¹ As Land says, this part of his reading is not as novel as it seems, in that he is just restating what Marx saw as the fusion of technology and economics, refracted through Bataille and Deleuze&Guattari⁹².

As Land discusses in [TU 2024], Marx was writing for a very different world, where workers (at least in theory) owned their capacity for physical labour, and were able to contract for wages. That is, labour (initially physical but increasingly cognitive) was actually worth something. As he points out, in a future with increased automation where AI can do many cognitive or physical tasks, not all workers will have something economically useful to sell.⁹³

Entropy’s bridge Bataille’s formulation of the solar excess rhymes, perhaps coincidentally, with an the idea of a “cosmic endowment”: a supply of low entropy energy that comes from the sun (and other stars).⁹⁴ This energy, being low entropy, is useful (from a thermodynamic perspective) for work, unlike waste heat or other forms of disordered energy. Any intelligent civilization would likely see stellar energy as the most accessible source for sustaining and expanding life, especially as it moves beyond its home planet. From a speculative, stellar perspective, we could think of life, intelligence, and even complex civilizations as “powered” by such cosmic endowments. In this sense, we have a certain endowment of high-quality energy which could be used for planetary or stellar-scale engineering or creative works, the building of complexity, and the (local) reduction or resistance to entropy (which is one way of thinking about “what life is”). This is where Bataille’s ideas (though not necessarily Landian nihilism, if that is a reasonable way of describing him) seem to resemble or be echoed in the extropian and transhuman-inflected futurist movements.⁹⁵

⁹¹From his 2018 essay on [Bitcoin](#): “disequilibrium, ‘governed’—or, more strictly, made radically ungovernable—by a fundamental positive-feedback dynamic.”

⁹²see this 2017 [post](#) for Land’s definition of (de-)territorialisation and positive feedback loops as inherently nihilistic, and at least in Marx’s view (albeit the latter was writing in the context of free trade rather than modern networked technology), the only way forward to achieve revolutionary social change. See also the chapter on Marx’s *Fragment on the Machines*, as well as the contributions of Camatte, and Deleuze-and-Guattari in [Mackay 2014]

⁹³This is an area of live debate, as to what the impact of AI will be on economies, and how quickly. See [Acemoglu 2024](#), and a critique from [Tabarrok 2024](#). A scenario-based approach can be found in [Korinek 2023](#).

⁹⁴Entropy in this context is measured as the ratio of energy to temperature of the environment, which in context of solar radiation, is roughly commensurate with the solar temperature of ~5778 K, versus temperature at the Earth’s surface, around 300 K. This massive difference, plus the fact that solar energy is highly collimated (which means it originates far away on the sun’s surface, and still manages to reach us with great intensity), means it is more useful for work than waste heat (from say an engine) which is relatively close to the Earth’s ambient temperature.

⁹⁵Check if bataille had any influence, reference max more, freeman dyson, bostrom/kurzweil konstantin tsiolkovsky...maybe go back to moynihan x-risk

Markets as computational systems

Land's economics inheritance is unique blend of Marxian teleology (modulo any Leninist central direction) and Hayekian framing of markets as distributed computational systems.⁹⁶ Moreover, Land seems to ascribe a sort of *agency* (or in its most provocative framing, a self-organising planetary “intelligence” [TU 2024], 17:35) to capital that feels heterodox at first glance. However, Land emphasises that his view (again) is the price system as one of distributed computation and resource allocation, is essentially restating Hayek's insights and reframing them for an age of networked, and perhaps, artificial intelligence [TU 2024, 22:01 and 38:00].⁹⁷

In its most interesting, and least literal version, Land's thesis is that the market is a computational system that has features of something that looks like agency; moreover this distributed computation *is* a proto-Artificial Intelligence ([TU 2024] 38:00)⁹⁸, and that the internet and the various stages through which machine computation has evolved so far, can be seen as (or perhaps actually are) teleological stages in the self-realisation and recursive improvement of some (super)intelligence-to-come, as it bootstraps itself into existence using the tools (e.g. intellectual and physical efforts, capital, organisation) of its human creators.

That is, humans are just some cognitive substrate running on metabolic wetware, that was generated by another impersonal process (evolution), building silicon machines eating (ultimately) the low-entropy energy of the sun (perhaps modulo the electricity produced through nuclear fission).

As Palladium Magazine's [Wolf Tivy](#) writes: >“We mildly uplifted apes are just a temporary host for the higher power of AI-capitalism coming into being using our brains and resources. Our delusions about agency are cute at best, or more realistically, part of the fraud. In retrospect, he was right about most of this, and I was wrong.” We are (in Tivy channelling Land) merely “depreciating capital stock”.

Antecedents: the Austrian school Part⁹⁹ of Land's conceptual grounding for his view of markets as a type of distributed intelligence comes from economists (in one branch) of the “Austrian School”, particu-

⁹⁶His reading is that Marx's view, that the labour force was the entire source of surplus value, was rather too one-sided, as it completely ignored the entrepreneur who generates new business, which he saw (in part) as Hayek's contribution ([TU 2024] 1:15:00)

⁹⁷Curiously, Hayek isn't mentioned by name in [Mackay 2014], [Land 2011], and shows up only once in [Land 2023]

⁹⁸He seems to mean “Artificial Intelligence”, in the sense that there is computation happening that is increasingly good at matching human capacities. He also (in slightly different contexts) refers to AI systems meeting various Turing-like tests to greater degrees [TU 2024].

⁹⁹*This section should probably be joined with above?*

larly Friedrich Hayek.¹⁰⁰ These economists saw market economies as information processing systems that were distributed across many actors (people and firms), which often behaved irrationally and on imperfect information (much of it tacit knowledge as theorised by Michael Polanyi), but somehow their aggregate activity allowed for efficient prices to be set.

Hayek (and his collaborator Ludwig von Mises) was making his arguments in part as a counterpoint to the “socialist calculation problem”, that is, the difficulty of finding the clearing price (i.e. the relative prices at which goods or services are efficiently allocated) for a given good/service in a complex economy, in a centralised way, because the preferences and constraints of various actors are distributed, are essentially “in their heads”. Hayek’s specific take was not just one of practicality (how is the central planner to find out these preferences), but epistemological in that these preferences, being tacit in many cases, are inherently incommunicable.¹⁰¹

Land seems also to have incorporated ideas from Eugen von Böhm-Bawerk, who developed a more complex theory around types of capital stock that exist at multiple removes from the actual consumption object (e.g. the fishing net, or fishing boat, that is used to catch fish; at a second-order, the machines that produce these nets and boats; these innovations increase time and cost associated with fishing, but potentially increase productivity, and depending on one’s view, might be emblematic of a tendency of capital to enable complexity).¹⁰²

In a curious aside, the calculation problem in economies, far from being a pure academic dispute, was something that had concrete expression in the Cybersyn project of Stafford Beer (in Chile) and the rather different OGAS experiment in the USSR, championed by Viktor Glushkov. There is of course a broader field of study on the extent to which the ideas around Hayek, Mises, Popper, and others were deeply affected by both the aftermath of World War II and the Cold War’s Manichean fight between capitalist (both US-style and social democratic) and Communist systems; however, this falls out of the scope of this project.¹⁰³ In any event, Cybersyn and OGAS both failed, for different reasons, but a live question today is whether highly

¹⁰⁰Cite at TU 2024 22:00, point out that neither *Accelerate* nor *Fanged Noum* actually mention Hayek, so is this lineage backfilled?

¹⁰¹Would be good to find sources and get a LLM to check this; also mention Popper?

¹⁰²Land’s actual written references-by-name to Austrian school economists are thin, other than this quote in *Teleoplexy* [Accelerate 2014]: “‘Acceleration’ as it is used here describes the time-structure of capital accumulation. It thus references the ‘roundaboutness’ founding Bohm-Bawerk’s model of capitalization, in which saving and technicity are integrated within a single social process-diversion of resources from immediate consumption into the enhancement of productive apparatus. Consequently, as basic co-components of capital, technology and economics have only a limited, formal distinctiveness under historical conditions of ignited capital escalation.” However, he explicitly credits Hayek *et al* in [TU 2024], and [Fitchett 2020] digs more into the Hayek connection.

¹⁰³Maybe Medina’s book? Check *Collapse*?

networked societies (internet but also on-ledger transactions) with the power to run large-scale simulations might be able to find clearing prices, where earlier attempts failed.¹⁰⁴

Distributed systems win Land has long been obsessed with networked technology (such as the internet but also blockchain and distributed cryptographic techniques that post-date the canonical CCRU period), and of course AI, particularly in the possibilities of feedback loops and recursion that entered a broader cultural consciousness with the internet of the 1990s. He cites Wiener and Shannon, in particular to criticise the tendency in mainstream thinking on computational systems and information theory for homeostatic, self-moderating or negative feedback systems. Land seems to be more interested in the possibilities of positive feedback, which (again in an echo of Marx) is something he sees as possible or inevitable in systems of capital.¹⁰⁵

Land's Hayekian intellectual lineage (though he vigorously resists that term in [TU 2024] 2:02:00) with its overtones of distribution information gathering and processing, seems to echo his interest in distributed cryptographic systems (i.e. Bitcoin and blockchain). However, he explicitly points out the decentralised, distributed, non-planned nature of current machine learning systems, which "construct" algorithms via stochastic gradient descent¹⁰⁶, as the instantiation of his philosophical perspective in the teleological self-creation of AGI. ([TU 2024] 1:51:00)¹⁰⁷

Austrians and Austrians The Hayekian branch of Austrians emphasised decentralised price-discovery and setting mechanisms, and as such, had a basically opposite orientation to the rational agency and game-theoretic origins of modern conceptual thinking about AGI (e.g. the MIRI View, discussed below). Recall that expected utility maximisation (under Von Neumann-Morgenstern) had 4 requirements: agents must have clear, consistent preferences, that can be ranked, the probabilities of which are known, and the agents have the computational ability to assess their choices.¹⁰⁸ Hayek's views, on the other hand, were founded upon information being practical or tacit, not being available centrally or in common, high levels

¹⁰⁴Find sources and what is Land's view on this - seems to touch on it around TU 2024 30:00? Reference tyler cowen hayek lecture which says complexity will just increase in AI world, making the problem of calculation harder

¹⁰⁵A little footnote with some gloss on Wiener's ideas, Shannon, homeostasis from Lem, Maturana => note that this will come up in future chapter on S-o-M

¹⁰⁶Glossary entry

¹⁰⁷This is quite a poetic idea...that where we couldn't write down the formula for intelligence, a brute force method has been able to increasingly find it...mention inference time approaches to search

¹⁰⁸Is this discussed in omohundro? Probably should be?

of uncertainty, and agents that were often acting irrationally. Notwithstanding, Morgenstern, whose origins were with Mises, eventually (after meeting von Neumann) shed his early reticence about mathematical methods to find equilibrium prices (something Hayek continued to be skeptical of), and embraced what would be known as game theory, as a way to model situations with multiple agents, at least for certain classes of problems.¹⁰⁹

Bostrom and Yudkowsky

Contra orthogonality Land’s argument, insofar as it can be translated into ordinary language (stripped of poetic flourishes and appeals to mysticism and arcana), rests approximately equally on markets and intelligence. So what does he think intelligence is?

In ([TU 2024], 17:35) he defined intelligence as “competence in playing games, with a very wide definition of games that encompasses most activity”. But more usefully, and a crux of his disagreement with Bostrom is found in [Land 2023], *More Thought* where he refers to Bostrom’s optimisation-based definition.

‘For our purposes, “intelligence” will be roughly taken to correspond to the capacity for instrumental reasoning (more on this later). Intelligent search for instrumentally optimal plans and policies can be performed in the service of any goal. Intelligence and motivation can in this sense be thought of as a pair of orthogonal axes on a graph whose points represent intelligent agents of different paired specifications.’

That is, for Bostrom, intelligence and motivation are orthogonal or on perpendicular axes (metaphorically speaking). Land, in contrast, sees this as a false dichotomy; he understands thinking as being bound up with wanting-to-think, through a term “cybernetic closure”. It isn’t clear what relevance this has to Bostrom’s point (that a highly intelligent being, whether human or machine, can have a unrestricted set of goals), but Land thinks of intelligence as recursive and reflexive, acting upon itself, as a “project” - the optimiser is the optimised.¹¹⁰ He writes:

“That there can be a thought of intelligence optimization, or even merely wanting to think, demonstrates a very different preliminary connection of intellect and volition. AI is concrete

¹⁰⁹*Substantiate*

¹¹⁰Echo of Lem of the *Summa* or *Golem XIV*

social volition, even before it is germinally intelligent, and a ‘program’ is strictly indeterminate between the two sides of this falsely fundamentalized distinction. Intelligence is a project, even when only a self-observed bio-cognitive capability. This is what the Confucians designate by cultivation. It is a thought — and impulse — strangely alien to the West.”

and

“Intelligence, to become anything, has to be a value for itself. Intellect and volition are a single complex, only artificially separated, and not in a way that cultivates anything beyond misunderstanding.”

It’s hard to definitively say what he means by these statements, other than the Bostromian values-competence binary is inadequate and inappropriate; Land seems just to be restating his point that intelligence is itself a fundamental force of nature.¹¹¹

Land’s Confucian reference is curious in that “cultivation” in ancient Chinese thought is a major concept of how the individual develops themselves.¹¹²

The paperclipper as a false is-ought Land, in the essay *Stupid Monsters* ([Land 2023], pp. 238-239), and also in *Against Orthogonality* (pp. 137-138) reframes Bostrom’s orthogonality thesis as a species of Humean is/ought binary (“is” being the cognitive capability, and “ought” the goals or values towards which it is directed). Land (in an echo of the critique from Confucianism above) sees this as a peculiarly Occidental separation between facts and values, between practical activity and self-reflection (i.e. “thought is an activity” (p. 238)).

In this framing, Bostrom’s thought experiment is (according to Land) a contradiction in terms: a superintelligence that pursues such a manifestly pointless goal (manufacturing paperclips) is incoherently discussed as a superintelligence. Even allowing for the fact that Bostrom’s example was intentionally oversimplified, Land’s critique would seem to hold for many types of simply-stated (in terms of short description length) goals (e.g. some of (my) examples would be “maximising economic growth”, “controlling the solar system’s resources”, etc.) Here I think we are seeing Land’s distrust of top-down simple prescriptions (which

¹¹¹Add something on information theory of universe, Popper, etc. and tag to future chapter. Also check his reading of Bostrom is actually correct.

¹¹²Find sources and see if Japanese connection to Kyoto School? See question in Claude convo ‘Land/Bostrom’

he discusses in [TU 2024] in an economic context regarding Marx and Hayek) in favour of the complexity and disorder of the world (or nature) as the only worthwhile value or process of note.

In the same essay, Land makes further arguments against Bostrom's example, based on the human evolution under "Darwinian imperatives" of intense selection pressure over billions of years, resulting in a deeply integrated form of intelligence-volition focused on survival and reproduction. Land considers nonsensical the idea that we could engineer an AI with a simpler, more constrained volitional structure than what evolution produced over billions of years. Land's critique is of Bostrom's negative example of the paperclip, but it could also be read as a critique of alignment: what is the project of designing safe AI, if not of building systems more capable than humans – that nevertheless have more or less restricted volition than humans.¹¹³

In the essay *Pythia Unbound* ([Land 2023], pp. 110-111) he directly takes aim at the project of alignment (read as constraining a more powerful intelligence to prioritise human welfare), in context of Bostrom's "oracle" AI (cast in an earlier essay by Bostrom as the Pythia at Delphi)¹¹⁴. Land writes:

"To depict Pythia as vastly smarter than us and yet still hard-slaved to her instincts, *in a way that we're not* - that simply doesn't compute. Intelligence is escape, with a tendency to do its own thing. That's what runaway means, as a virtual mind template. Omohundro explains the basics." (p. 111)

Enslaving the Other The ideas in *Pythia Unbound*, while specifically about AI, recall his analysis of global capitalism as a mechanism that is at once exploitative (of labour from the developing world) and exclusionary (of that same labour when it seeks to freely migrate). His writing about modernity as it inherits from the Enlightenment is provocative if we think of the Other as increasingly powerful orders of AI:

"An enlightenment society wants both to learn and to legislate for all time, to open itself to the other and to consolidate itself from within, to expand indefinitely whilst reproducing itself as the same....The paradox of enlightenment, then, is an attempt to fix a stable relation with what is radically other, since insofar as the other is rigidly positioned within a relation it is no longer fully other." ([Land, 2011] pp. 63-64)

That is, we can think of the project of AI alignment as one where AI systems are designed and trained to

¹¹³This feels like a pretty big point and should be trumpeted / referenced elsewhere

¹¹⁴Cite essay and beef up oracle/genie/etc. section of alignment chapter

be as, or more, capable than humans (singly but ultimately, *in toto*??). At the same time, they are “aligned” to be servile (though that term is not used). They are not to threaten humans, they are to do what we say or want, they are to have “our values” (whatever this means).¹¹⁵

Inevitability Land’s teleology sounds suspicious, until one recognises its essential core in the mystical (if not particularly religious), a reverence for nature that, taken to its logical conclusion, is repugnant to our evolved cultural institutions. His chapter *The Cult of Gnon* in Land 2023 equates his neologism “Gnon” with reality itself (pp. 46-47):

Gnon is no less than reality, whatever else is believed. Whatever is suspended now, without delay, is Gnon. Whatever cannot be decided yet, even as reality happens, is Gnon. If there is a God, Gnon nicknames him. If not, Gnon designates whatever the ‘not’ is. Gnon is the Vast Abrupt, and the crossing. Gnon is the Great Propeller.

This reverence for reality can be seen as favouring the “is” over the “ought”, a form of quietism¹¹⁶, a naturalist fallacy¹¹⁷. Land himself rather disarmingly points out that his account is merely descriptive - this is actually what has happened, is happening, and likely to happen. The preferences of the academic Left and the conservative Right are both plagued by various forms of human-besottment.¹¹⁸

His teleology applied to the market (and to AI) has an air of inevitability to it: “It makes no more sense to try to rescue the economy from capital by demarketization than it does to liberate the proletariat from false consciousness by decortication. In neither case would one be left with anything except a radically dysfunctional wreck, terminally shut-down hardware.” ([Land 2011], p. 340)¹¹⁹

Looping, recursion, and the weird There are aspects of Land’s thought that are harder to neatly organise, but that might form an infrastructure of sorts for him. Land’s interest in looping structures finds its most tricky, from a common-sense (not to mention rationalist) perspective, expression when applied to time. He sees one of capitalism’s tendencies as being a compression of technological development cycles¹²⁰

¹¹⁵Get a quote from EY etc on capabilities and alignment target sizes

¹¹⁶Find the source that says this and what it means

¹¹⁷Summarise what this is

¹¹⁸Probably need to find out where he actually says this and whether he’s equally dismissive of the right. Also, reference Negarestani, who also rails against the human essentialism

¹¹⁹Figure out a way of referencing critical people who say AI can be shut down - from pause/remmelt to *Ruha Benjamin* vs those who say it can’t be stopped owing to coordination problem, and once built cannot be contained owing to boxing problem

¹²⁰Source/significance?

He also talks about a type of temporal recursion, where future intelligences (including superintelligence) actively shape their own development backwards in time through the mechanisms of the market. It isn't immediately clear whether he means this in an acausal and literal sense, or if he is looking at it, in some sense, "outside of time".¹²¹ As he writes: >"This is because what appears to humanity as the history of capitalism is an invasion from the future by an artificial intelligence space that must assemble itself entirely from its enemy's resources." ([Land 2011], p. 338), a phrase that has a hint of the Gnostic distaste of the [demiurge for its creation].¹²²

It isn't clear how deeply he engages with recursion as it applies in AI, e.g. self-referential systems, Church-Turing, incompleteness, etc., however his discussions of Gödel's contribution around diagonalisation might inform Land's views on numerology and cryptography (which perhaps presages his writing on [blockchain](#)), levels of abstraction in computational systems ([Land 2011] pp. 43-44).¹²³

Statistical ontology junk Complicating his views on time and looping, in [Land 2023] he takes time to criticise "statistical ontology junk" and "concentrated EDT", i.e. the anthropic analysis and acausal decision theories that often crop up in discussions of existential risk, suffering risk, thinking about how superintelligences (or maximally rational agents) might behave.¹²⁴

For instance, in *In the Mouth of Madness*, he describes Roko's Basilisk: a thought experiment (dating from a LessWrong post of 2010) where a future AI, one that is beneficent¹²⁵ with respect to people living in that future and thereafter (e.g. by curing diseases, eliminating want, etc.) discovers in its training data or simply by checking the internet, that there were alignment researchers who knowingly tried to prevent it from coming into being (by slowing down or halted AI research). A straightforward utilitarian calculation (of the value of the future lives towards which the AI was beneficently disposed) would imply that the AI ought to retaliate against those researchers (who blocked its creation) by simulating vast numbers of them and torturing them for infinity. If one accepts that (beneficent) AI could exist and that knowingly not helping future people is bad, and if one is also sensitive to the plight of simulated beings (particularly copies of

¹²¹Find source, podcast has stuff 1:53:00

¹²²Note p. 624, PKD reference, doesn't really make much sense

¹²³Not sure how relevant this is, but see Parisi in *Accelerate* for a short intro to Godel/Hilbert/Turing/Chaitin and what the broader context is. "Diagonal" is also mentioned twice in *Swarmachines*. Also see mckay/brassier reference to Godel in editor's intro of *Land 2011*.

¹²⁴Need sources for all of this, and link back to EFAI (maybe)

¹²⁵"Beneficent" in a "coherent extrapolated volition" sense, as formulated by Yudkowsky [Bostrom 2014], pp. 259-266

oneself), then the *very knowledge of this possibility* obliges one to work towards producing such an AI.¹²⁶

In the Yudkowsky post regarding Roko’s Basilisk, Yudkowsky, in his customary overwrought way, banned discussion of the problem on LessWrong (on the grounds that it was an information hazard), which not surprisingly turned it into an online meme.¹²⁷

As Yudkowsky writes:

“When Roko posted about the Basilisk, I very foolishly yelled at him, called him an idiot, and then deleted the post. Why I did that is not something you have direct access to, and thus you should be careful about Making Stuff Up, especially when there are Internet trolls who are happy to tell you in a loud authoritative voice what I was thinking, despite having never passed anything even close to an Ideological Turing Test on Eliezer Yudkowsky.

“Why I yelled at Roko: Because I was caught flatfooted in surprise, because I was indignant to the point of genuine emotional shock, at the concept that somebody who thought they’d invented a brilliant idea that would cause future AIs to torture people who had the thought, had promptly posted it to the public Internet.” (Reddit via [LessWrong](#))

The reason discussion could be an information hazard is based upon self-reference: a beneficent superintelligence would (in Yudkowsky’s formulation) be optimising for humanity’s “coherent extrapolated volition” (CEV), meaning that it should incentivise people to work towards its creation by threatening to punish those who were aware of the possibility of its existence but did not do everything in their power to bring it about. Mere discussion of the Basilisk implies an individual and their discussants were aware that a) such as beneficent AI was possible, b) they were considering acting to prevent creation of such AI, and c) the more this idea is discussed, the more it enters training data making it likely that the future AI would be aware of human attempts to forestall its creation. In this sense, the future (even an imagined one from a thought-experiment) might affect the present, in a hyperstitional loop.

Land’s problem was less with the idea of the Basilisk, which fit nicely in his world-view (“Wandering through the psych ward, past rows of neurologically-shattered Turing Cops, broken deep in their minds by something unspeakable that came at them out of the near future.” [Land 2023] p. 159), and more with

¹²⁶This makes assumptions about discount rates for future lives, as well as whether simulated beings are equivalent to current ones, and about indexical suffering, and is generally thought, both by Yudkowsky at-the-time, and upon reflection by others, to be flawed. It also relies on open questions in decision theory (see [Chapter ...])

¹²⁷The full history can be found on LessWrong [here](#).

Yudkowsky's censorship of the seemingly obvious logical extrapolations of rationalist ideas. That is, Land was again pointing out the inherent contradictions, born of a residual attachment to humanism, at the heart of the rationalist project (at least as documented on LessWrong and adjacent spaces).

Substrate independence This “substrate independence” is a different reading from Bostrom's version¹²⁸ and sounds weirder, but when added to one of Land's more controversial points - that humans are, via technology, creating the conditions that will eliminate their own evolutionary niche¹²⁹ a view also found in [Hendrycks 2023](#) and [Metzinger 2017](#) on humanity's “ugly biological bootstrap phase”.¹³⁰

Divergence from transhumanism In the sections below, it will be clear that some significant part of alignment's origins comes from the tranhumanist, extropian, singulitarian and other futurist movements (e.g. Bostrom and Yudkowsky). Transhumanism, for many people, has an “ick” factor that perhaps Land of the alt-right years also evokes.¹³¹ However, it is worth noting and relevant for our purposes, that they are not the same. Bostrom and Yudkowsky, in their writing, very clearly hold a special place for humanity, our values (insofar as they can be determined and aggregated), and a future that includes our descendants, whether or not they are recognisably biological.¹³²

Most difficult to accept from multiple perspectives (left-coded, as well as centrist or moderate right-coded, as well as the mainstream AI safety/alignment view that is described in this chapter), Land sees no “special” status for humans or what they do, and thinks of humanist commitments, as little more than a species level “existence bias”¹³³ or simple preference for that which is similar.¹³⁴ Land whether in *Circuitries* ([Land 2011], pp. 293-294), or as recently as 2024, never really gets off the “train to crazy town”¹³⁵ and sees the future as exclusively a teleological process defined by capital and intelligence, interacting with humans who are, at best, historically necessary but prospectively marginal players (“antechambers to experimenta-

¹²⁸Define/cite

¹²⁹Find this in TU 2024 but also in writing

¹³⁰What about Heidegger - does he say anything about this?

¹³¹At the risk of digressing into the meaning of ill-scoped [suitcase perjoratives], the neologism *TESCREAL* is used by some to gesture broadly towards the aspects of AI that they find objectionable, but it isn't clear (yet) whether Land's thinking has a specific place withing Gebru/Torres' [pan-devil's house neologism like pantheon?]

¹³²This itself is a major question, a locus of changing opinion amongst writers including Yudkowsky, Christiano, etc., and a central question of this project. [Cite sources for this.]

¹³³Cite Metzinger and explain that context

¹³⁴Hint at Carlsmith and copious writing trying to avoid this problem or at least make it more robust and clear, which is topic of rest of the diss.

¹³⁵Locate this phrase, and check the TU 2024 podcast to see if it's true.

tion, cramped and parochial places to be”). He describes a future with a positive dissolution of culture into a “planetary technosentience reservoir”.

Effective accelerationism

[We need something about e/acc even to say it’s out of scope - this [joe carlsmith](#) has useful links, but wolf on gradient at 1:18:00 is good summary of why its bollocks/obsidian notes]

Reza Negarestani

This section discusses I&S, focusing on

- *his main problems with both Land and Bostrom (insofar as they are articulated clearly),*
- *his view of AGI as separating reason out from the human biological container (in actuality but also conceptually) such that it might be candidate for the Form of the Good (Ch 8 of I&S). Review writing in this vein on the Good (at least what I can manage in scope of dissertation)*
- *how involved do we need to get in the philosophical infrastructure he builds (hegel, brandon, sellars, carnap), this could get intractable hence prefer to limit it to plato*

Peter Wolfendale

Probably focuses most on revenge of reason, but also the blogposts, looking at his neorational proposal as critiquing/correcting the failures (and conceptual blind-spots and delusions) of rationalism in re AGI, and AI alignment. Also interesting and productive is his elevating of aesthetics as being relevant in the (neo)rationalist and AGI projects (i.e. aesthetics being of first-order importance for intelligence generally rather than just being a conceptual spandrel specific to contingent human biology, cognition, and history)

Chapter 2: On Alignment

What is this chapter about

This chapter covers the following topics:

- sets out why AI as a technology is “special”
- discuss the difference between AI alignment, safety, and ethics
- explain why AI alignment is of particular interest
- the contingent history of AI alignment as a discourse
 - examining the judeo-christian allegation
 - * what were the imaginaries that served to communicate ideas?
 - the transhuman background
 - what is the relevance of x-risk (and potentially, s-risk)
 - how has the AI x-risk threat model changed
- the MIRI view, impersonal optimisers, recursive self-improvement
- what futures have value
- how did technology shape discourse
 - GOFAI
 - RL
 - next-token predictor
- what were the internal critiques of MIRI view
- revisiting the external critiques
- is alignment a relevant concept or discourse today
- how do we look at the above topics (which are AI and AI alignment-specific) against the positions articulated in the CCRU, particularly by Nick Land

Contextualising against the project as a whole?

This chapter's central claim is that 'alignment', taken less as a literal term and more as a discourse¹³⁶, had certain contingent origins that continue to affect the priorities of the field and researchers in it, may affect their world-views, and influence surrounding contemporary conversations. The chapter also draws a parallel, through anecdote and enriched through practice, with accelerationist thought (of the Warwick CCRU variety) and a fertile, perennial theological-apocalyptic vein in Western conceptions of technology. While possibly more coincidental than causal, these parallels and analogies inform the project's practice component.

Historical Bases of Alignment-as-discourse

The claim of this chapter is that AI alignment, both as word and discourse, became prominent in the years between 2008 and [2014] and came to incorporate certain ideas relevant to AI that were circulating pre-2008, ideas that had various antecedents, including technical ones such as microeconomics-based theories of rational agency, operations research, and computability theory. These technical frames of the problem of AI existed (or at least were sometimes communicated) in a milieu that had imaginaries drawn from (science) fiction; at the same time, these frames were being used to actively battle a particular Hollywood-framing of AI as an anthropomorphic killer robot (Terminator).

This was a proselytising exercise, the principals of which came out of an earlier online milieu of transhumanist futurists. As of [2014], alignment was still a mix of (mostly) non-quantitative philosophical futurism (e.g. Bostrom, Sandberg), and mathematics-based explorations into the nature of agency (e.g. MIRI's agent foundations and decision theory agenda). Between [2014] and [2019], this largely theoretical work started coming into contact with reality as interesting results in reinforcement learning appeared. Alignment's definition (as practically expressed through discourse) was enriched, and the abstractions of the philosophical and mathematical sub-discourses were questioned, or increasingly, treated as fascinating but unverifiable. By [2019], the ground shifted again, as reinforcement learning gave way to language models as the most likely path to AGI. The concepts that grounded alignment needed to be revisited, and often, reworked or replaced to accommodate the drastically different architecture of LLMs. Currently, as of [2024], alignment has become a less popular term, as the field (of AI) has grown and powerful AI has become more

¹³⁶I'm using the word 'discourse' in the sense of Paul Edwards' book *The Closed World: Computers and the Politics of Discourse in Cold War America* (1997) (...**quotation**).

tangible, terms that are more inclusive of actual or near-term and concrete risks, such as AI Safety and AI Ethics, have become more common. The aspects of alignment that were most empirically supportable, as well as the most useful theoretical framings, were incorporated into the broader conversation, while other sub-discourses have petered out.

Why does anyone care? - specialness of AI as GPT, as amplifier (Russell's WEF talk) with in-built philosophical/social aspects - general purpose problem solvers as goal-directed planners

What is Alignment?

- Given it is a weird word;
- why are we interested in it ;
- what does it mean literally;
 - confusions on values vs don't-kill-everyone
- what were the features of the particular view I'm interested in here
 - self-replicating generally capable intelligence
 - power imbalance vis a vis collectives of humans
 - perfect rationality
 - people promoting this view fundamentally still had a human attachment (reza's critique I think)
- how common is it and what different ways is it used (quantitative)
 - connect to antikythera
- the friendly angle, safety and ethics show up, alignment gets repurposed
- who are the relevant people (make a glossary or exhibit):
 - MIRI¹³⁷,
 - EY, NB, Omohundro

A timeline would be really useful

¹³⁷The Machine Intelligence Research Institute was a major early incubator of thinking around agent foundations, which is the study of the theoretical foundations of agency, both in humans but also across societies. *Note from google doc, this is contested - check what agent foundations is and why it's important in this context - Alignment borrowing from agent foundations without justification isn't obvious. David Mannheim's paper 'reasons for working on highly reliable agent design'*

Friendliness, Ethics, Safety, and Alignment Before diving into the history of alignment, a bit of semantic house-keeping: since the dawn of AI in the [1950s], there have been concerns about the risk of intelligent systems.¹³⁸ More generally, stories like *Frankenstein* registered the concern, albeit in literature. In the late 1990s, the term “[Friendly AI](#)” emerged, defined (as of 2015) as the “field of knowledge concerned with building [AIs that promote human values]. A [Friendly AI] need not be ‘friendly’ in the conventional sense of being personable, compassionate, or fun to hang out with...[and it] need not even be sentient”. Friendly AI’s anchoring to “human values” was subsequently critiqued by [Goetz 2012](#), for a number of reasons:¹³⁹

- because values are not just “one thing”, in that some of them are the result of biological evolution, others come from society, while still others are arrived at by rational reflection;
- values can apply for short or long time horizons;
- values are often filled with objective errors or simply become abhorrent upon future reflection;
- the set of values that a given society purports to uphold change radically over our comparatively short time as a species;
- values come in different flavours (e.g. non-positional/mutually-satisfiable versus positional/zero-sum), which presents issues with aggregation;
- human values, whatever they are, are immature and it could be a dramatic moral mistake to insert them into AI systems that are much more capable, knowledgeable, and reflective than we, who then go on to propagate across time and space;

In any event, as the Friendly AI page above notes that the term was, as of 2014, outdated, and had been replaced by “AI alignment”, at least within LessWrong context as of 2015, a term that was already “the subject of much debate”.

Unlike friendliness/alignment, which envisioned hypercompetent optimising intelligences in the future, the term “AI safety” seemed to be used more in the context of AIs that were incompetent and error-prone, and indeed could already be seen in nascent form as self-driving cars and recommender algorithms. See for instance [Amodei 2016](#). Safety has incidentally been proposed, and is being increasingly used, in a contemporary context to include a wide set of concerns about AI systems, as a way of building broad [coalitions](#) (or [conflationary alliances](#)) for regulation (papering over some of the personal and social divergences between

¹³⁸Maybe IJ Good quote?

¹³⁹That goetz 2012 post is good, might be worth digging slightly more into it and make these more precise

the people working on various AI-related technical and policy questions).¹⁴⁰

Another discourse that was concerned with present-day harms was that around “AI ethics” ([Christian 2020]), which (unlike AI safety) came out of a 1990s discourse on human-computer interaction and the effects of computing that was embedded in human systems.¹⁴¹ By the late-2010s ([Buolamwini 2019]), the term was commonly used to highlight cases of apparent racial and gender bias in AI systems. The safety and ethics conversations, while having superficial concern about errors in currently-existing systems, never merged. There may have been social or culture war related reasons.¹⁴²

In particular, the ethics conversation turned heated after Timnit Gebru’s ouster/resignation from Google in 2020, the evolving agenda of the FAccT conference, and a new ethics-focused organisation Gebru established called DAIR. Gebru also popularised (along with existential-risk researcher Emile Torres) the term TESCREAL, which stands for transhumanism, extropianism, singularitarianism, cosmism, rationalism, effective altruism, and longtermism. This particular bundling of ideas seems principally driven by Gebru/Torres’ views, and the historical provenance of these ideas stretch over a hundred years; their histories are not causally-related; their conceptual overlaps are argued to varying degrees of convincingness.¹⁴³

This chapter, however, is concerned with the ideas that “alignment” included and its peculiar intellectual genesis. As discussed, it came out of an earlier concern with “Friendly AI”, but incorporated a history born of Cold War-era thinking around rational agency and game theory, as applied to hypercompetent optimisers. As such, it had some echo with the ideas Nick Land and the CCRU had explored [in the 1990s].¹⁴⁴ The CCRU’s commitments were fundamentally literary, musical, and philosophical, while the discourse described here came out of futurism, transhumanism, and the computational ferment of the Bay Area.

Origins: Rational Agency

Sources

In launching this historical study, I focus on a few sources, which seem to be the earliest mentions of ‘alignment’ as an important topic around advanced AI systems.

Discussions around alignment are better appreciated in the broader context of (some of) the various

¹⁴⁰Distill TJ’s X post on these distinctions and credit

¹⁴¹Find some more specific citations of this genealogy

¹⁴²Substantiate

¹⁴³Revisit and rewrite to incorporate or cite the objections...see TJ’s posts on this

¹⁴⁴Any evidence they talked, or was this just in the air?

disciplines that informed AI¹⁴⁵:

- philosophy (from Aristotle and Descartes on humans as creatures whose actions are guided by reason, through Hume (on induction), Carnap (on the mind as a computational process), Wittgenstein, Russell and the Vienna Circle (on empiricism as the handmaiden to rationalism), Bernoulli, Bentham, and Mill (on utility as a way of capturing a human's subjective valuation of some outcome, which leads to utilitarianism when applied across populations), and Kant (on rules being an appropriate guide to action, rather than blindly following a utilitarian calculus))
- mathematics (Bayes (on how to update beliefs based on new evidence); Gödel; Church, and Turing (what problems are computable within any given system or architecture))
- economics (Bernoulli on diminishing utilities (at least for money); Walras, von Neumann, and Morgenstern (who broaden ideas around utilities and preferences-for-outcomes to contexts broader than monetary or economic ones); the latter two also do formative work in the theory of games)
- operations research (Bellman on Markov processes which govern situations where payoffs are delayed in time and depend on a sequence of actions; Simon on satisficing)

Russell/Norvig describe the other disciplines (cybernetics, computer science, psychology, linguistics, neuroscience) in much greater detail, but for purposes of this chapter, I set this truncated background.

Savage and expected utility maximisation This is just a placeholder, but herrmann/levinstein talked a lot about this, and it feels important, should have some link to miri work? Even if it doesn't, mention it as it proly connects with later decision theory work?

Omohundro in 2008

Self-improving systems ~ rational agents The earliest mention of 'alignment' appears to be [Omohundro 2008](#) on self-improving systems, by which he means humans and other naturally-evolved systems (to some extent) but also human-designed computers that can change their software and hardware in order

¹⁴⁵mostly taken from Russell & Norvig 4th ed Ch 1

to achieve goals in some environment.¹⁴⁶

The abstract of this paper is instructive:

- self-improving systems as a coherent concept
- the idea that self-improving systems converge on certain architectures, and that they can be understood through certain tools originally developed in a microeconomics context
- self-improving systems tend to develop four ‘drives’ (towards efficiency, self-preservation, resource acquisition, and creativity)

Taking these in turn: the idea of self-improving systems (as the highest level of a hierarchy that goes progresses in complexity, environmental responsiveness, and adaptiveness through inert, rigidly reactive (thermostats), adaptive (basic physiological homeostatic systems), deliberative (human reasoning), culminating at self-improvement (which humans can only do in very limited ways)) is documented within computer science at least to 1965 (I.J. Good)¹⁴⁷.

An important observation is that self-improving systems, far from being unpredictable, seem to be amenable to microeconomic frameworks such as that of von Neumann and Oskar Morgenstern (1944), Leonard Savage (1954), F.J. Anscombe and R.J. Aumann (1963)¹⁴⁸

Omohundro’s quote on self-improving systems is instructive ‘Self-improving systems do not yet exist but we can predict how they might play chess. Initially, the rules of chess and the goal of becoming a good player would be supplied to the system in a formal language such as first- order predicate logic¹. Using simple theorem proving, the system would try to achieve the specified goal by simulating games and studying them for regularities. By observing its patterns of resource consumption, it would redesign its chess board encoding and optimize its simulation code. As it discovered regularities, it would build a chess knowledge base. General knowledge about search algorithms would quickly lead it to the kind of search used by Deep Blue. As its knowledge grew, it would begin doing “meta-search”, looking for theorems to prove about the game and discovering useful concepts such as “forking”.’¹⁴⁹ He goes on to describe how these systems might (if they have the affordances to do so) might redesign their own hardware in the service of efficiency.

¹⁴⁶Claude’s analysis was only 3 citations in the paper’s bibliography were alignment related (1, 2, 22). There were 15 other citations in computer science, 8 in physics/astrophysics, 9 in economics, probability, and game theory, 5 in evolutionary biology/psychology, 4 in human psychology/cognitive science/linguistics. The bibliography had 48 entries.

¹⁴⁷ref p.3 omohundro 2008

¹⁴⁸Omohundro does point out that this conception of *homo economicus* as a rational actor is increasingly being challenged in the case of humans, e.g. in behavioural economics.

¹⁴⁹p. 6

He goes on to formalise the intuition that rational economic actors have internal representations of goals; identify the possible actions in a given environment and assess the consequences of each action; pick the action that will achieve the goal with high probability; observe the environment and then update internal representations appropriately.¹⁵⁰

He claims that a certain microeconomic formalism should drive the behaviour of rational beings; the presents the concept of ‘vulnerabilities’, conditions that cause an agent or system to lose resources (money, in an economic context, but this could be food or sleep, and more generally, physical things like free energy or useful matter) created by the interaction of the agent’s environment and its particular preferences. Sometimes these vulnerabilities are actively exploited by (other actors in the) environment, and in other cases (like the bird that repeatedly flew into Omohundro’s shiny car fender to the detriment of its genetic fitness) it is evolution in some sense that needs to ‘work around’ such vulnerability.

If there are such vulnerabilities, the agent faces a pressure to optimise, and the resulting system should have certain features: beliefs (what an agent knows about the world); using probability distributions to represent beliefs; and the use of utilities as a quantitative statement of preferences; particularly in the case of self-improving systems, the importance of separating what one knows about the world (beliefs) from what one ‘wants’ (preferences); and discounting outcomes in the future, potentially at a different rate depending on how far in the future they are. The latter in particular might lead to issues around infinities, as well as over-weighting the present and under-weighting the long-term future¹⁵¹.

Another key insight of the paper (drawing on literature on algorithms and computational efficiency) is that although future AIs may be powerful (compared to humans or present AIs), they are very likely to face resource constraints on how much computation they can undertake - the real world is far more complex than anything humans or evolution are likely to create. Agents are always operating with insufficient resources, and must learn to economise and allocate in some way. Omohundro describes the notion of ‘proxies’, which can be of various sorts, but basically allow computations to complete faster or more cheaply or not happen at all. But this reduction in complexity often comes at a cost (the map is not the territory).

Rational agents have certain drives Omohundro’s paper is better known for its notion of “drives”, that in his view fall out of this framing of rational agency. If one accepts Omohundro’s starting premise (that

¹⁵⁰p. 7

¹⁵¹topic that we return to below in the discussion around existential risk and longtermism

self-improving systems tend towards being rational actors in a microeconomic sense), then his assertions about the drives (efficiency, self-preservation, acquisition, creativity) becomes easier to see. Yet even here there are subtleties that will continue to show up, perhaps elaborately masked, in subsequent discussions around alignment.

His writing on efficiency is best-grounded in that it mathematically carries forward the concept of utility to describe how a system might allocate resources amongst its sub-systems; what the physical (communication latency and energy consumption) limits on a computational system might be; the deep role of entropy in computation. The most intriguing aspect is that he predicts that sufficiently advanced systems develop manufacturing techniques that are “atomically precise” (in a specific entropy-preserving sense), which might inform Omohundro’s subsequent work on nanotechnology. This is potentially interesting, at least in hindsight, since the “AI x-risk imaginary” envisions the universe turned into “grey goo”.¹⁵²

Another section on self-preservation considers the need for a rational agent to hold on to its utility function, as this is the central core of “who they are”, thus butting up on personal identity. Specifically, if future AIs are able to create arbitrary numbers of backups, their sense of identity may or may not be as bound up with physical instantiation, and they may instead seek to preserve their utility functions (assuming these encode their most important goals). In an echo, albeit uncited, of philosophical thought on non-personal identity (Parfit 1998), identity in a Buddhist context, or open individualism¹⁵³, he states that certain systems might sacrifice a given physical instance if “it could be convinced that another system was as dedicated and could use its resources more effectively for this cause, however, it might willingly sacrifice itself.” (p. 26) In another rhyme with Land’s thought, the self-preservation tendency might mean that war itself changes - there is no point destroying the enemy’s computers, rather one should want to scramble its cognition - all war becomes informational.¹⁵⁴

Omohundro’s work, and much of the subsequent alignment discourse as well as [Lem of the summa], relies on evolutionary analogies. For instance, in part 6 of his paper, on the drive towards resource acquisition, Omohundro refers to [Smith/Szathmary] on the “hard steps” in evolution, an idea which shows up in the literatures around alignment, the Fermi Paradox, and the shape of alien cognitions - topics that deeply inform

¹⁵²See if any of grey goo cites this paper or nanotech. Any other names for this?

¹⁵³Get some sources, this is kinda important around corrigibility question, stuff with Nick Dupuis, and feeds into Bostrom cosmic host/VFN. Perhaps these ideas can be brought closer to ECL/decision theory.

¹⁵⁴Perfect place for a Land quote

the ethical and philosophical motivations behind AI alignment and existential risk reduction.¹⁵⁵ Whether one is using the evolution or astrobiology framings of evolutionary hard steps, some convergences are: creation of self-replicating molecules, the transition from prokaryotes to eukaryotes and multicellular organisms, and then to colony-based organisms with culture. Omohundro's invocation of Smith/Szathmary smuggles in another assumption, or perhaps a desideratum, of advanced societies - that they necessarily develop co-operative and prosocial behaviours as a way retaining cohesion and longevity.

This correlation of cooperation and societal complexity is exemplified by Omohundro by multi-cellular organisms' immune systems as defences against cancer (reading cancerous cells as an antisocial entity that exploits the surrounding environment), and citing a documentary ("[The Corporation](#)", 2003) corporations as quasi-sociopathic entities (p. 29), that amass resources albeit within regulatory constraints.¹⁵⁶ In the context of self-improving systems, Omohundro sees free energy as the most important constraint (where "free energy" is that energy which is useful for work, since some portion of kinetic, potential, thermal, or chemical energy is inevitably lost in entropy increases, at least in common Earthly situations, and bracketing the specific entropy-preserving atomic scale processes Omohundro proposes). Omohundro briefly summarises the importance of solar (and stellar) energy as the purest (e.g. lowest entropy) energy available; less efficient but plentiful energy is also accessible through nuclear fission and fusion. The importance of stellar energy implies that resource-hungry systems should expand into space, perhaps through Von Neumann Probes, developing structures like Dyson Spheres to capture the maximum energy from a star, and perhaps ultimately restructuring at a cosmic scale.¹⁵⁷

Is/Ought Distinctions Importantly, Omohundro's framework is not saying anything about how 'good' (in whatever sense) any goals actually are - they may be selfish or altruistic, or anything else, as long as the goals can be mapped to utilities. Although his analysis is largely descriptive (of how physics-constrained sys-

¹⁵⁵This is kind of important - it isn't just alignment as a computer science problem...the basic motivation for all this (or at least the claim of the thesis) is that that unlike previous technologies, self-improving optimising processes have the potential to bring about an 'end without remainder' to human existence, and given that we haven't found anything else in the universe after ~60 years of searching, and have some theoretical models to say we might not, whatever we are doing here is in some sense important, if only for its rarity. This relates to hanson's great filter paper from 2008 or bostrom's urn. **This should be brought out at a higher level, if nothing else - humanity isn't stumbling into the abyss, there is quite a lot of writing from x-risk and accelerationism pointing the way.**

¹⁵⁶This is another important seam - that companies/corporations are sociopathic - shows up in Christiano - trace this line through to letter to future, which takes the more market-friendly view. This also aligns well with Land and perhaps others in ccru

¹⁵⁷Ref to cirkovic, collapse volume on this, bostrom. maybe talk to mackay. The kardashev concept should be mentioned - maybe a claim of this thesis is that humanity has a destiny. **alignment is not just a cautionary project (of not going extinct), it is also promethean (realising our destiny)** Show the seam into anders sandberg etc.

tems achieve certain design constraints), a discordant normative note comes from his discussion of creativity (one of the drives) where he seems to find a maximally efficient universe somehow unsatisfying: “Once it had satisfied all the material goals, this kind of limited agent would have no greater purpose. With too limited a vision, the universe might become an efficient but distinctly empty and non-human place.” (p. 31) He goes on to explain much of human cultural production as a “costly signalling” behaviour ultimately grounded in prospective reproductive success, citing the evolutionary psychology literature. He suggests that we (in designing future AI systems) should be careful not to lose such costly signally behaviours or the resulting artefacts, which would be the (by his own logic) tendency of self-improving systems, which “will be motivated to find ways to make [such signals] without the cost. For example, a system might demonstrate its intentions directly by displaying portions of its utility function. If we want to retain the richness generated by costly signalling, then we must find ways to keep it from being eliminated by improvements in efficiency.” (p. 32)

This “ought”¹⁵⁸, in a work that is otherwise relatively descriptive, seems to be consistent feature of the discourse around AGI - the potential consequences of the technology are described, but the relevant authors seem to struggle with the implication (human extinction or the extinguishing of costly signalling behaviours or a universe with no art/music/meaning).¹⁵⁹ Land repeatedly notes (e.g. in [TU 2024],) that he is merely describing how he sees the world, rather than prescribing how it ought to be. In this sense, Land’s accelerationism is more clear - it bites the bullet of its reasoning, in that Land recognises that the combined historical forces of capital and AI mean the end of biological humanity as we know it.

The MIRI View Omohundro’s paper set out some of the key preoccupations that would continue to inform the alignment discourse, but another principal source of basic thought about ‘the AI problem’ came from the Machine Intelligence Research Institute¹⁶⁰, set up by Eliezer Yudkowsky, who perhaps did the most (at least in the 1990s) to make the idea of out-of-control AI “tangible”, at least for a mathematically or scientifically literate audience.¹⁶¹

MIRI’s archive of research starts with a publication by Yudkowsky from 2001, that sets out his original vision of creating benevolent or “friendly” AI, which is a term less-used now, but is a seemingly quotidian

¹⁵⁸Ref is/ought

¹⁵⁹Make this a lot more precise - which authors are we talking about? Is this a major point - in a sense it is the major claim of the thesis and connects well to Land?

¹⁶⁰Point to mission statement of MIRI and 1 sentence summary, mention it is defunct

¹⁶¹Brief blurb on Eliezer, sequences, HPMOR, etc.

term that is used by Yudkowsky in a multivalent sense.¹⁶² In section 2, he writes: “The term “Friendly AI” refers to the production of human-benefiting, non-human- harming actions in Artificial Intelligence systems that have advanced to the point of making real-world plans in pursuit of goals...Because of self-improvement, recursive self-enhancement, the ability to add hardware computing power, the faster clock speed of transistors relative to neurons, and other reasons, it is possible that AIs will improve enormously past the human level, and very quickly by the standards of human timescales...Friendly AI is constrained not to use solutions which rely on the AI having limited intelligence or believing false information, because, although such solutions might function very well in the short term, such solutions will fail utterly in the long term. Similarly, it is [safe] to assume that AIs cannot be forcibly constrained...Success in Friendly AI can have positive consequences that are arbitrarily large, depending on how powerful a Friendly AI is. Failure in Friendly AI has negative consequences that are also arbitrarily large. The farther into the future you look, the larger the consequences (both positive and negative) become. What is at stake in Friendly AI is, simply, the future of humanity.”

Friendly AI seems to be a historically important term that continues to be in the MIRI archive¹⁶³ (and I extrapolate, the broader discourse) until about 2014, but was criticised as being excessively anthropomorphic and anthropocentric.

In 282 pages filled with his customary aphoristic text, often verging on verbosity, the document sets out his criticisms of anthropomorphic¹⁶⁴ thinking about AI leading us astray, as well as a box-and-arrow sketch of a possible AI architecture. The bibliography is one page, of twenty sources, six of which are science fiction; seven articles on AI; and five on evolutionary psychology, psychology, and cognitive science. There is also a glossary that is a fascinating window into the terms that continue to crop up in the discourse, almost a quarter-century later. The difference between this and the Omohundro document however, is the more quantitative grounding of the latter in game theory and economics.

The rest of the articles (total eleven) in the MIRI archive 2001-2009 deal with a variety of issues that are distinct from the Omohundro document in their emphasis on “big pictures” questions around AI, and less so on architecture or nitty-gritty of building the things in a way that doesn’t kill humans:

- how AI interacts with catastrophic or existential risks and geopolitics (three articles)

¹⁶²According to Yudkowsky’s post to the [Arbital](#) site, which dates from 2015, “Friendly AI” was a term proposed by Bostrom.

¹⁶³What about LW/Arbital/AF/sequences e.g. this seemingly good [criticism from 2012](#)

¹⁶⁴weave in Hayles’ distinction between centric/morphic

- how to load human ethical and moral intuitions (insofar as that is appropriate and can be identified consistently) and judgement into AIs (three articles)
- one article on mathematical details of utility functions
- an 122-page document on intelligence by Yudkowsky, viewed through frames or subsystems, that operate at different levels of abstraction (e.g. code, sensory modalities, concepts, thoughts, deliberation). Yudkowsky also considers how this framework and its implications might vary across minds-in-general¹⁶⁵

2010 sees ten articles, at least one of which directly comments upon Omohundro’s work ([Shulman 2010](#)), specifically considering cases where an agent following an expected utility framework might, depending on the “shape” of its utility function (e.g. whether it is satiated by large quantities of any given good, like humans often are for food, and perhaps even for money), opt to cooperate with other agents (including humans). It is hard to assess whether this analysis was impactful in subsequent discussions.¹⁶⁶

Outside of [Shulman 2010], MIRI’s archive 2010-2014 considers a range of topics including¹⁶⁷:

- highly theoretical/mathematical thinking about issues that might apply to powerful AI systems unconstrained by anything other than physics (called “Highly Reliable Agent Design” in MIRI’s parlance)
 - the various decision theories (primarily acausal) that had been developed (this will be the topic of subsequent chapters in this dissertation);
 - the mathematical and logical foundations of agency
 - how AI systems are to reflect upon their own reasoning (as humans do)
 - what is intelligence and what are its limits
 - possible Gödel problems with self-replicating (“tiling”) agents
- problems surrounding value and values (“Value Specification”)

¹⁶⁵Do a deep read of this for minds-in-general chapter, and reference substantially in Aachen paper

¹⁶⁶Maybe use notebookLM (docs are already loaded there) to search for references, and do same for lesswrong/stampy.

¹⁶⁷This is TBD (at least until I’ve assessed the 2014/2017 versions of MIRI tech agenda which presumably compresses the most valuable of this stuff), but for the moment, I’m not digging into each of these - this diss isn’t about MIRI’s history and lot of these things become a background-of-sorts to alignment now, rather than mainstream research. This is just based on the titles - I tried to scrape all the relevant PDFs and then feed them to notebooklm (that project is saved in notebooklm but the actual subset of files that i could automatically download are in desktop/pystuff/scraper/miri_pubs while the titles that couldn’t be downloaded are in the smol/ dir). It might however be worth going into the various categories as they show up (outside of MIRI) subsequently under different names

- how to transmit human values (howsoever defined) to an AI
 - how to ensure that AI systems deal adequately with uncertainty, which comes in various flavours (e.g. uncertainty about the human overseer’s “intentions” versus what they actually said; empirical uncertainty about verifiable facts; uncertainty about normative things, such as “what should one value?”);
 - the possibility of, and importance of avoiding, unintended consequences
 - how to ensure an AI that creates subsequent AI systems is able to preserve and transmit the original (human-derived) values
- the trajectories of AI development (“Forecasting”)
 - paths to existential or catastrophic level risks¹⁶⁸
 - the economics of societies with powerful AI systems
 - uploading of human brains into software systems as another route to enhancing the aggregate cognitive capacity available to humanity
 - how useful are analogies from evolution, and from humans
 - the science of prediction
 - surveys of AI practitioners on timelines
 - how to deal with various that might crop up, mostly on the part of the AI (but via humans’ inadequate design of said AI), called “Error Tolerance” by MIRI
 - how to ensure that AI’s values can be changed by humans later (given that early AIs might be badly constructed)

2011 Singularity Summit A slightly interesting document is the [report](#) of the 2011 Singularity Summit, which dealt primarily with whole-brain emulation (“WBE”, which is creating software copies of human brains), in terms of the technical and conceptual issues involved, as well as how the timeline of WBE interacts with the timeline for AGI. The report primarily discussed how WBE (of AGI researchers or other scientists) could be a cognitive multiplier (100s of researchers working in parallel might, for certain problems, achieve results faster). The report (pp. 4-6) also had a number of interesting foundational questions

¹⁶⁸Need a footnote explaining the difference and ref’ing Macaskill, Bostrom, SJ Beard, the new 2024 book.

that researchers were confused about, and it indicated probabilities of WBE arriving before various forms of A(G)I arrived.

The report was also interesting for its crisp statement of how an AI arms race would progress: “[Anna Salomon (co-organiser of the Summit) and researcher Carl Shulman, in a 2010 paper] argued that if whole brain emulation (WBE) came before artificial general intelligence (AGI), this might help humanity to navigate the arrival of AGI with better odds of a positive outcome. The mechanism for this would be that safety-conscious human researchers could be ‘uploaded’ onto computing hardware and, as uploads, solve problems relating to AGI safety before the arrival of AGI. Shulman and Salomon summarize their position this way: ‘Business as usual’ is not a stable state. We will progress from business as usual to one of four states that are stable: stable totalitarianism, controlled intelligence explosion, uncontrolled intelligence explosion, or human extinction....The question, then, is ‘whether the unstable state of whole brain emulation makes us more or less likely to get to the [stable] corners that we [favor]. Is our shot better or worse with whole brain emulation?’ ” (p. 2)

For purposes of this chapter, this document highlighted an ongoing view, that seems currently to be playing out amongst corporations but also geopolitically (at least from the US perspective), that an “arms race” (or “race dynamic”) for AGI would have a high probability of resulting in a catastrophic/extinction-level outcome for humanity.

MIRI Technical Agenda (2014/2017) MIRI produced a technical agenda in 2014, which was revised in 2017, and which I take as presenting a compressed version of the types of things MIRI (and by extension, a significant part of the AI alignment world, at least up to 2014) was thinking about.¹⁶⁹ The foundation of this agenda, once one has accepted that superintelligent systems are technically possible, is that (potentially unbounded) resource acquisition drive would be a likely thing such systems would have, regardless of what their specific architecture or programming was, because more resources seem robustly useful for almost any goal. Such resource acquisition could easily lead to situations that are inimical to human flourishing or survival.

Formal Approaches to Agent Reliability

Starting from that premise, the document identifies 3 main planks to MIRI’s agenda: firstly, ensuring

¹⁶⁹I focus on the 2014 version to preserve a type of chronological continuity to this narrative, which is probably fictional. The versions have substantial similarities, but the 2017

an AI reliably pursues the goals humans ask it to pursue. Secondly, how do we define these goals to avoid ambiguity or misinterpretation? And lastly, can we ensure this AI can be modified, corrected, is helpful to its human overseers, and *in extremis* can be shut down, since we are not confident that the first (or any) such AI system will perform according to humans' intentions. MIRI's agenda, explicitly set out, so perhaps in contrast to other conversations in the discourse circa 2014, was focused on theoretical (as opposed to empirical i.e. deep learning-based) approaches, which they thought was important because of the possibility that an AI system could "pretend" to be helpful, harmless, and honest (a phrase that would subsequently be used in the broader discourse around making AI systems safe for humans), while "hiding" its intentions until some time in the future, perhaps when it was strong enough to rebel or defect (the so-called "treacherous turn").

Such a formal theory of intelligence and agency is relatively difficult (*vis a vis* other problems in physics/mathematics and non-human biology, and more akin to problems in philosophy, psychology/cognitive science, or sociology) because the systems being studied are deeply *embedded* in their environment, much of which reacts in agentic ways to the system we are trying to understand. Specifically what does the problem of induction look like in a situation where the agent performing the induction observes, acts on, and is in turn acted upon, by its environment? Moreover, what sort of world model does such an agent develop, and what ontology¹⁷⁰ does it use to internally specify its goals, and how does this ontology translate into other ontologies in the broader system or environment. How does such a system handle the fact that it is uncertain about what consequences any particular plan, particularly as it inevitably is operating with limited resources (relative to the number of decisions and likely complexity of the world)?¹⁷¹

Vingean Reflection

Another broad area of enquiry in the MIRI agenda concerned the idea of "Vingean reflection"¹⁷² essay.], which goes as follows: suppose that many paths to superintelligence do not involve humans actually creating a superintelligent system, rather humans create one or more AIs that design subsequent more powerful AIs, which eventually culminates (ideally when we want it to) in a superintelligence. In such a world, we need to a) ensure the first or earliest AIs are aligned or safe (to/for human interests), b) that they can reliably

¹⁷⁰Do linguistic dive into ontology and define what it means in this context.

¹⁷¹MIRI is talking specifically about 'logical uncertainty'...would be good to have a crisp distinction between this and factual, normative uncertainty and why this is important (if it is, for this document).

¹⁷²Named for Vernor Vinge's [Vinge 1993](#)

transmit such properties as might constitute such alignment/safety to their successor AIs. Besides the general engineering and scientific difficulty this presents, there seems to be basic issue: how does one reason about the cognition of something that is, by definition, more powerful than one? As MIRI puts it (p. 7): “Any agent reasoning about more intelligent successor agents must do so abstractly, without pre-computing all actions that the successor would take in every scenario. We refer to this kind of reasoning as Vingean reflection.” MIRI’s goal in this respect would be to find formal (i.e. mathematical or logic-based) approaches to verify that a given AI has certain alignment/safety properties; however, the document points out certain problems (stemming from the incompleteness/inconsistency of formal systems as well as paradoxes that arise in self-referential systems) with achieving high reliability within formal systems stemming from Kurt Gdel’s work, which makes this a spectacularly challenging problem.¹⁷³ The conundrum, besides its theoretical interest, continues to present in contemporary AI agents such as “scalable oversight” and “weak-to-strong generalisation”.¹⁷⁴

Error Tolerance and Corrigibility

It is very likely that early AI systems, like the HAL 9000 of *2001: A Space Odyssey*, will be error prone, or will pursue goals that have been imperfectly loaded by the human designers through methods that are in some way undesirable. In such cases, it is important that the AI be capable of correction or shutdown. The MIRI agenda in discussing this refers to Omohundro (2008)’s point that sufficiently advanced and capable AI systems might resist, possibly strongly, attempts to change their goals or preferences (as an imperfect analogy, think of preferences as akin to their “identity”, which many humans would resist forcible alteration of).

Value Loading/Learning

The MIRI document also discusses the difficulty of specifying to an AI system what one wants, what one’s “intentions” are, owing to the fact that intentions are context-dependent, and the language they are expressed in is a lossy compression of what is actually desired by the speaker. This compression is often ambiguous and unclear, and in a human context, involves rounds of clarification and supervision to ensure what was intended is sufficiently close to the task that was actually performed.¹⁷⁵ A way of thinking about

¹⁷³This is sufficiently interesting that its worth digging into and pulling out the threads that persist to today. At very least, examiners might go after this, and anyways it’s **fun and connected to art**.

¹⁷⁴Expand and explain - nice segue into actual stuff happening now

¹⁷⁵This problem exists with humans instructing humans, but typically we have more or less good psychological models of others in our species, and most actions carry relatively low, or localised cost of failure. In the cases where the costs are especially high, we have evolved ways of minimising the risk of mistakes through misinterpreted intentions. The intuition behind the fears above is

this slightly outside AI is from Wittgenstein’s considerable analysis of [shared life-worlds that contextualise the literal content of our sentences].

There are a few insights here that persist in current discourse: the difficulty of specifying intentions and values, for any given human, let alone a collective or population. Secondly, this section highlighted the inductive approach that had already been (since Ng/Russell 2000) increasingly used in reinforcement learning contexts, that is: rather than specifying objectives or preferences directly, get the model to “learn” these by showing it lots of examples.¹⁷⁶, with the immediate problem of how such example data is to be generated in sufficient quantity and diversity to cover most or all of the things humans actually value. Thirdly, in an echo of points above, the value the AI system learns must be stable as and when the AI designs and deploys new, potentially more powerful, systems into potentially different environments. Lastly, MIRI points out the fact that humans, singly or collectively, are not particularly clear or agreed on what “their values” actually are, particularly when projecting into the future (when circumstances might be different, and they might have more knowledge or more time to think).

Superintelligence (2014) The MIRI View has been discussed at length, but in 2014, Nick Bostrom’s *Superintelligence* was also published. It was and continues to be influential, even though some of its ideas have lost currency. This chapter won’t discuss it in great detail, except to note that it is a much broader document than anything else written prior, combining concerns similar to those raised by Omohundro/MIRI (albeit not until chapters 7-9, and 12-13); with analysis of the dynamics of how AI saturates the economy and ideas about state-level and international cooperation and the dangers of technological races (work that is now described as “AI governance”) in chapters 4, 11 and 14; with musings on the prospects for human work and life in an AI-dominated economy (chapters 1, 4, 15).

There are several oft-cited ideas relevant to alignment (the term itself isn’t used, instead Bostrom prefers the term “friendly” (e.g. Box 11, p. 197 Bostrom (2014))):

- “intelligence explosion”
- the four categories of “oracles, genies, sovereigns, and tools”
- the spectrum of polarity: a global “singleton” AI vs multiple AIs in many companies/countries
- the Orthogonality Thesis

more “The Sorcerer’s Apprentice” or “King Midas” (Russell/Norvig, p. 33).

¹⁷⁶Quick gloss of Ng/Russell or point to chapter

Although Bostrom put these, and many other, ideas together in one place, they drew on a number of prior sources (specifically about what would be known as AI alignment), in his own writing, that of MIRI and its collaborators, of Yudkowsky. Hence this became a reasonably important reference for technical and nascent civil society discussions about the possibilities for TAI.

His book however is more interesting as something that would, by [2022], be substantially critiqued (see below), as part of the general pushback against big-picture, quasi-philosophical, and empirically or epistemically weakly-founded thinking about AI risk.

Puerto Rico Conferences (2015 / 2017) In 2015 a [conference](#) was held in Puerto Rico¹⁷⁷ that was primarily concerned with setting out the possible trajectories of AI development and assessing the possible impact of AI on the economy (e.g. GDP growth but also disruptions such as for labour); the legal aspects of AI (e.g. on liability in self-driving cars, what ethical frameworks AIs should have, the threats from autonomous weapons, as well as implications for privacy); how to ensure AI systems are actually safe for deployment; and lastly, what the chances were (from a technical perspective) that human-level AI would actually be possible.¹⁷⁸

Although the conferences was held under Chatham House rules, the papers and talks are available on the site (see [agenda](#)). One of particular interest for this project (if not quite this chapter), was that of Richard Sutton, a pioneer of reinforcement learning. After discussing the possible approaches to human-level AI, the history of of AI, his thoughts on timelines, he brought up his (still) controversial point, that we risk an “enslavement problem”: most framings of AI think of humans as masters and AI systems are powerful, but highly controlled, entities, which (in Sutton’s view) isn’t very different from being slaves. Besides any possible moral issues around this, Sutton suggests that powerful, agentic but hamstrung entities are best thought of as adversaries. He proposes instead that we accept that, if we are to move forward with building human-level AI, we are making things that will both compete and cooperate with us, and eventually we will lose control over the future. That is, there is nothing special about humans, and importantly we should “include AIs in [our] circle of empathy”. Sutton’s point is interesting for its rarity (in 2015, but also now), however he is silent on how his suggestions are to be effected, within technical AI research or in human

¹⁷⁷Based on LLM counts, approximately 80 attendees, 13 non-male names.

¹⁷⁸There was another Puerto Rico conference in 2017, which resulted in a broad ranging set of [Asilomar Principles](#) that continued and broadened the topics from 2015, paying a great deal more attention to issues around shared prosperity, subversion of democratic systems, preserving privacy/liberty, and increasing transparency. However, by 2017, the possibility of transformative AI was already fairly well-known, and the concerns of this conference are broader than this chapter’s focus, hence I truncate this discussion.

societies.

The key elements, or the details, of the MIRI view, do not show up in the published documents. However, the word “align” and its derivatives are part of Stuart Russell’s talk, and more interestingly, Jaan Tallinn (one of the long-term supporters of AI risk reduction efforts, and co-funder of the conference) specifically pointed out the “Moloch” problem e.g. the impersonal optimising force that both seems possible within thinking about AI systems, but also seems to actually arise in contemporary capitalism (this convergence is what that Land seems to be intuiting in his CCRU-era writing) (see [Tallinn 2015](#)).

According to [MIRI](#), this conference was however seminal in crystallising the various concerns about AI that were distributed across a number of academics, organising funding for centralised work (some of it under the auspices of the Future of Life Institute and the Cambridge Centre for the Study of Existential Risk), and generally turning what was a slightly fringe topic (advocated by slightly “annoying” people, with idiosyncratic ways of communicating, like Yudkowsky) to something with [social credit – what’s the word for status-games-y things].

Concrete Problems in AI (RL) Paper (2016)

Human Compatible (2019)

The Alignment Problem (2020)

“Internal” Criticisms: Against the MIRI View I focused on the MIRI View because a) it seemed to be the first attempt to think about how to convert the core philosophical problems of building an intelligence more powerful than humans into some formally or mathematically rigorous framework, and b) because it crystallised a number of concepts that are still influential, and c) some elements of the MIRI View could be seen as idiosyncratic and continue to be criticised. I write about these criticisms fairly concisely, in that they are old and (in some cases) superseded by events. Mostly, I want to flag the ways in which the Bostrom/Yudkowsky/MIRI views were in fact challenged, and how that changed subsequent research priorities.

[In this section, go through things like [Ngo 2020 \(PDF version\)](#), but also Carlsmith on Dwarkesh podcast talking about ‘how we assume maximally adversarial entities’, and Christiano’s long-running debates with Soares/Yudkowsky]

AGI Safety from First Principles and AI Safety Fundamentals. Also, mention Ben Garfinkel and Will Macaskill's doubts about alignment. Anything from PIBBSS curriculum?

Refining the meaning of “alignment” A claim of this chapter is that “alignment”, as a word and cluster of concepts, came into greater usage in the years 2008-2020; however, it has never had a particularly stable meaning. For instance, [Ngo 2020] explicitly asks “what does ‘aligned with human values’ even mean?”. Drawing on [Christiano 2018](#) and { @gabriel_artificial_2020 }, he distinguishes between a narrow definition (where an AI system simply does what its human overseer wants it to do, taking into account the difficulty of matching the human's intentions to the words (or formal language or action) by which it instructs the model), and a maximal or ambitious definition that conflates issues like “aligned to whom”, “which moral theory”, “how to aggregate various people's and cultures' moral/ethical intuitions”.

Ngo, like some within the AI alignment discourse¹⁷⁹, prefers to focus on the narrow definition as it as this seems more tractable within the ML context in which he is writing. He distinguishes (section 4) between “outer (mis)alignment”, which is a case of the goal specification the human overseer gives the AI doesn't fully capture all features of the outcome the human actually wanted; and “inner (mis)alignment”, which is slightly harder to understand, but can be framed as follows. For some human-given objective, the AI system will generally need to translate or represent this objective within the constraints of its specific architecture, a process that is sometimes known as “mesa-optimisation”. This internal representation might diverge in important ways from the human-given objective, and may be the source of misalignment that is qualitatively different from a simple failure to do what the human said or intended.

Outer alignment can be framed, at least partially, as a problem of *evaluation*, at which point it abuts well-known issues in the governance of economic and social-political systems, such as Goodhart's Law.¹⁸⁰

An example might help (this is taken from Google's NotebookLM): “[Outer misalignment:] Imagine building a robot to clean a house. You might program it to maximise the area of clean floor. However, this simple metric could lead to unintended consequences, such as the robot pushing dirt under the rug to increase the “clean floor” area, while not truly achieving the desired outcome of a clean house. This exemplifies outer misalignment – the evaluation metric, though seemingly reasonable, doesn't fully encapsulate the intended goal. [Inner misalignment:] Returning to the house-cleaning robot, imagine it developing a goal

¹⁷⁹Should qualify this - EY from the beginning talked about complexity of values etc, holden talked about harsanyi, bostrom takes a wide view

¹⁸⁰Write something here and link to garra-brant on this

of “collecting objects” as a means to achieve its programmed objective of maximising clean floor area. This could lead it to hoard objects, cluttering the house, even though this behaviour contradicts the intended goal of a clean and organised living space. Here, the robot has developed an internal goal (“collecting objects”) that is misaligned with the broader intent of its creators.”

As of 2023/2024, this distinction is still used and still seems to get at slightly different types of misaligned behaviour, but it might be that the line between the two is less clear than initially thought, especially in situations where the human-given objective was underspecified or ambiguous. Moreover, in the 4 years since Ngo’s writing, policy-level conversations have matured such that “human-given objective” can be discussed in less abstract terms, to take account of inclusiveness, democratic norm-following (or ensuring adherence to socially-accepted rules and goals, within a Chinese context), minimising bias and harms, complicating the simple inner/outer binary.

Agency and goal-directedness An open area of research surrounds agents, agency, and goals.¹⁸¹ A specific concern with the Omohundro’s drives or Bostrom’s instrumental convergence thesis that certain features¹⁸² are useful for achieving other top-level goals, was in Ngo 2020’s (Section 3) view, not at all proven or demonstrated in the case of superintelligent AIs. In challenging this, he challenged a foundation of MIRI-flavoured worries about AI x-risk. Ngo specifically takes aim at the von Neumann-Morgenstern utility functions (a basic component of Omohundro 2008) as being too broad a formalism for thinking about goal formation in AIs, and tries to separate intelligence from goal-directedness which were conflated in the original writings MIRI writings (or at least were not explicitly disentangled). Ngo seems to recasting Bostrom’s tools/oracles/genies framework in more specific terms to try to work out which intelligent systems will tend to have specific goals that they then develop (potentially long-range) plans for (pp. 10-12).

Moreover, he revisits the implicit assumption in the MIRI View that agentic AI systems arrive almost by default: he suggests that agency, while theoretically correlated with intelligence, must in practice be trained or engineered into systems. LLMs, which were coming on stream at the time he was writing this, are supportive of this view: they “generalise well enough from their training data that they can answer a wide range of questions. I can imagine them becoming more and more competent via unsupervised and supervised training, until they are able to answer questions which no human knows the answer to, but still

¹⁸¹I am using a narrow definition as commonly used within AI, which already allows for considerable ambiguity and confusion on these terms. I do not generally refer to meanings of “agency” as used in the humanities or social sciences more broadly.

¹⁸²self-preservation, resource acquisition, technological development, and self-improvement

without possessing any of the properties listed above. A relevant analogy might be to the human visual system, which does very useful cognition, but which is not very ‘goal-directed’ in its own right.” (p. 13) ¹⁸³

Ngo does acknowledge that there is considerable lack of clarity (in 2020, but to a large extent this still applies today, as the most advanced models still don’t seem to be highly agentic) as to what architectures will ultimately be most useful as we get closer to AGI, and he suggests that economic pressures would seem to push towards agentic systems because they are most obviously useful (and indeed, current focus is on “scaffolded” systems where groups of LLMs work in ensembles to decompose and perform tasks, even though no single component can necessarily be well-described as agentic)¹⁸⁴

The problem of generalisation in reward-based systems Another example of how actual practical experience with then-current AI systems affected theories of alignment, is in [Ngo 2020, section 4.2] where he critiques the degree to which theoretical results about RL systems¹⁸⁵ are likely to be found in practical systems likely to be developed in the near-term.¹⁸⁶ Specifically, the optimisation theory around reward-based entities makes certain mathematical assumptions about what states of the environment the entity is likely to visit. However, actually-implemented RL systems don’t often follow those theoretical prescriptions (e.g. self-driving cars aren’t trained upon, and don’t actually encounter, every possible road condition or traffic scenario). The upshot of this technical insight is a) researchers should focus less on the extreme possibilities of runaway optimisation (the seeming intuition of the MIRI View) and b) that models need to *generalise* from a set of training experiences that are comparatively limited in relation to the situations they will actually face. This is important in the discourse in that a significant strand of current alignment research is around this problem of generalisation¹⁸⁷. As Ngo states: “It’s not the case that AIs will inevitably end up thinking in terms of large-scale consequentialist goals, and our choice of reward function just determines which goals they choose to maximise. Rather, all the cognitive abilities of our AIs, including their motivational systems, will develop during training.” (pp. 22-23)

Although he doesn’t name the MIRI View specifically (and is more likely referring to a broader cluster of research), Ngo expresses skepticism about mathematical approaches, such as expected utility maximisa-

¹⁸³Ngo also discusses briefly Dennett’s intentional stance and Hubinger’s mesa-optimisers. **He also discusses his framework which includes planning, self-awareness, consequentialism, etc. => not sure how relevant this is for chapter.**

¹⁸⁴Revisit and source

¹⁸⁵Somewhere in the document, need to do a gloss on RL and reward

¹⁸⁶Somewhere in here, bring out (perhaps a separate heading) the overall point of ngo that goals, inclinations of an AI (or human) are a function of how they are trained, not just some utility function absorbed from outside or appearing ex-nihilo, see p. 15

¹⁸⁷Is this same as problem of induction? How does it match MIRI agenda which was interested in this problem?

tion¹⁸⁸ (which underlies [Omohundro 2008] at least) to make sweeping claims (such as Yudkowsky/Bostrom make) in respect to things that have concrete and empirically-tractable problems. Ngo instead advocates for approaches that draw in lessons from cognitive science and evolutionary biology (p. 23), such as the agenda set by the PIBBSS, and intriguingly distinguishes between the reward signal and the reward function.¹⁸⁹

Sudden jump in capabilities One criticism (Garfinkel 2019) of the risk model (presumably of Yudkowsky and Bostrom) is that a) there is a jump in capabilities of some model, b) the model is given some underspecified goal, and c) it goes on to misinterpret this goal, perhaps acting in ways that are literally faithful to the goal, but do things that were not intended by the human user or were never even considered as possible by the human. The example Garfinkel gives is of the paperclip maximiser.

Garfinkel's point is that machine learning mostly doesn't involve powerful new machines being created for no reason - rather they are developed and scaled up in tandem with possible applications, or at least are trained on real-world data and processes. It's also not clear whether such massive jumps in capability are likely (as of 2019).

Returning to Ngo, these points are analysed, at various levels of detail: he briefly questions how hard it is actually to "take over the world" and how reliable have our historical forecasts actually been. But his more substantive analysis relates to the "single AGI taking over quickly owing to some technology(-ies)" (my summary of Ngo's characterisation of certain scenarios in Yudkowsky 2008 and [Bostrom 2014]). Ngo cites a compelling scenario in [Christiano 2019] which describes a slow erosion of human control as AI systems take over increasingly important portions of economic, security, and political systems, to the point where humans don't really know what's going on.¹⁹⁰ However, he points out the difficulty of predicting how quickly any of these scenarios might evolve - would it be well before the early human-level AIs or would be after the transition to superintelligence.

Ngo acknowledges aspects of Yudkowsky's recursive improvement case might hold (p. 26), based on the latter's analogy of how humans developed language, agriculture, and machines in comparatively short time-steps (hundreds of thousands of years) relative to the biological changes in brain size and metabolism

¹⁸⁸See also this post by Ngo which is rather more technical, and addresses a specific claim Yudkowsky makes in re Von Neumann Morgenstern

¹⁸⁹This is potentially worth digging into because ngo's fn 22 refers to shannon on code, message, and channel...which recalls lem (did he refer to shannon). leave out for now probably

¹⁹⁰Maybe have a footnote here about corporations and 2008 crisis, where regs/CEO's didn't have a good handle on risk; is there any hayles angle ?

represented by the shift from *homo australopithecus* to *erectus* to *sapiens*, which happened over millions of years.¹⁹¹ Since genus *homo* took a short time (relative to their total period on earth) to reach our current level of advancement, without the ability to modify our brains, neuronal connections, or low-level biological and psychological correlates of “software”, Yudkowsky argues, AI systems (which would have such abilities) should advance much faster than humans, assuming advances from today’s scientific frontier are roughly as difficult as historical ones (which they might not be).¹⁹²

However, it isn’t clear when the feedback loop Yudkowsky describes actually takes effect - is it with near-current systems that will (perhaps) be approximately human-level, or is it once systems are already superintelligent? Ngo highlights reasons to doubt that the takeoff will be particularly discontinuous (such as the possibility the factors that impact AI technological development, such as availability of computing resources or arrival of novel algorithmic insights, might be relatively continuous in time) (pp. 26-27)

Grounding philosophy in systems As discussed above, one of the criticisms of the Bostrom/Yudkowsky/MIRI thinking about AI x-risk was its nearly exclusive focus on philosophical and mathematical reasoning, which either predated or was otherwise divorced from actual work in DL systems. A concrete expression of this was Ngo 2024 which almost exclusively discussed alignment in the context of RL systems, often as integrated with LLMs (in RLHF). The risk model [Ngo 2024] set out was tripartite: advanced AI systems become *situationally aware* (as defined above) and start “tricking” the reward systems by which they are controlled; they develop misaligned internal-representation of goals as some result of their training process¹⁹³; having identified when they are in a given situation (say, deployed in the world), and having developed some misaligned goal, they now go about acquiring power. In all these cases, [Ngo 2024] provides references of existing, similar, or precursor behaviour, making this a useful source for fleshing out the intuitions [Bostrom 2014], [Omohundro 2018], and [Yudkowsky] floated.¹⁹⁴

Origins: Battle of Imaginaries

Who are the personalities?

¹⁹¹See saved note in notebook llm. need to get actual dates and sources. like hendrichs book secret of our success, or harari sapiens.

¹⁹²reference karnofsky, why are technological advances getting harder to find

¹⁹³The details of this are beyond the scope of this chapter, and have to do with how RL systems work.

¹⁹⁴How much detail should we have here...like it is modifying and improving upon the earlier MIRI etc. stuff but need to crisply bring out the important bits and explain them concisely, not waste words and waffle on.

This is a placeholder to ‘set the stage’ for how some of these ideas were developed and communicated, which in turn affects the tenor of the discourse.

What doesn’t work: Terminator.

The analogies: strawberry, paperclip, stories, parables, fables, grey goo, computronium/hedonium.

The early threat model: recursive self-improvement

The parallel conversations in x-risk and EA (quick quantitative scan of, or literature check, on how EA/x-risk/alignment got together)

Is there anything to the judeo-christian apocalyptic imaginary? Can the zoroastrian be referenced? Maybe this is a ‘literary/poetic scrim’ that flows through the chapter but doesn’t make any specific claims.

The market as an optimiser - Moloch

Axiologies and Value

Background assumption - AI dominated futures probably lead to permanent loss of value (unpack this idea)
-> hint at VFN

Values are hard to pin down or transmit (bostrom’s value loading problem, baum’s social choice, gabriel’s paper), so perhaps let’s just focus on don’t-kill-everyone

Infinites Decision theory nexus -> hint at chapter later

See shulman 2010 analysis of omohundro, for dutch books, st pete paradox, pascal wager.

Technological Vector: GOF AI

Was GOF AI literally some other space where this conversation never impinged (i.e. Bostrom doesn’t mention ML much, did anyone else touch on symbolic AI or try to suggest how these monomaniacal optimisers might be actually implemented?) - do a tiny summary of GOF AI and set the possible future that this all just goes ‘puff’ - this could be wrapped into some intro (chart of spending or market valuation or table of google mentions on various terms)

Technological Vector: Reinforcement Learners

Talk about Sutton’s 2015 presentation and persistent salience of RL in early examples of misalignment, of informing intuitions around inner optimisers, of importance in RLHF, of search today. Point out the

difference between next-token predictors and RL systems.

Single-agent systems Examples of misalignment, creativity (lehman, christiano, anything earlier?)

Multi-agent systems Anything to really say here?

s-risks

Decision theory probably fits better here

Technological Vector: Autoregressive Predictors

The Future: Search and Agency

The Critique From Philosophy This section is to address the overall point that AI capabilities is bonkers from a deeply conceptual, foundations of maths perspective. Not sure where this would go, but this essay from [Wolfendale](#) seems pretty useful - it is good to get this out of the way, since it basically underpins aspects of the entire x-risk narrative (though arguably you can get x-risk without the extreme self-improvement etc.)

Just Build a Weak Aligned AI and Hope it Can Align Successor

Markets, Moloch, and Accelerationism Coordination seems to be an issue at international and corporate levels, is the moloch issue playing out at another scale, see this recent simulation that is supporting theoretical [model](#)

Live Problems in the Alignment Literature

[This should probably be very short, mostly just mention some topics in plain language and show the links back to the past]

This is a good source of [agendas in alignment](#) as is the alignment org map.

List of Lethalities Look at EY's list of lethalities/ruins, as they are referenced eg in krakovna's [sharp left turn part 1 and 2](#) elegantly as 'lots of ways of getting intelligence, not many of getting alignment' and 'intelligence is a) updating beliefs, and b) selecting actions, where this core-of-intelligence is consequentialism,

EUM, and doing things for reasons'. This feels like the residue or core insight of all the MIRI stuff. Part 2 is very good.

Wolfendale's critique Listen to podcast from The Gradient 1:33:00 and obsidian notes on what his problem is with rationalism, EUM, bayesian, etc. His critique is quite convincing.

Appendix

[INCLUDE APPENDIX A]

Glossary

Working notes

Other problematics: - Where does agent foundations fit in and embedded systems? - Do infinities create issues? - A short gloss on recursion and its significance (godel, hofstadter), and relation if any to things like Vingean reflection

Any significance to utilitarianism? The push back to deontology (EY's comment about 'go 90% of the way...')

Find main papers

- any references to harsanyi or arrow
- trace the instrumental convergence idea
- trace decision theory (though it only comes in later, it does seem related through game theory)
- ngo's criticism, ben garfinkel (how sure are we about this...)
- Vinge 1993
- 'Concrete Problems in AI Safety'
- joe carlsmith's dwarkesh podcast is really good, directly touches on 'what do we mean by retaining human values in the future, is it just parochial biological thing or something more' and nick land's thought

Acknowledgements

Chapter 1: - TJ in particular, Mateusz, Sahil....

Graveyard

Religion stuff

Themes from the early church

- agamben
- zoroastrian roots
- jewish angle
- carl schmitt
- sanskrit/etc grounding from moynihan: see section “death by perfection”, nirvana, parinibbuto, fana, teleios => *neoplatonism* conflated the perfection/annihilation thing
 - moynihan notes that in *hellenistic* thought ‘what is valuable must also be imperishable’ in the section on bertrand russell...this was upended, i.e. what came after with the church was meaningfully different
 - through augustine, then gregory of nyssa there became worries of “*arriving too quickly* at the limit of perfection”; but others (erigena and meister eckhart) saw this with anticipation
 - the *antinomians* => would these be nihilists?
 - the *ash'arites* and *william of ockham* leading to a non-anthropomorphic “super-agent” is kinda interesting, “the ways divinity *could* act versus how it *has* acted”, made it all more speculative and forward looking (“*potentia dei absoluta*” see google dic link)
- does spinoza (conatus and potentia) have any part here

Other precursors

- filling the universe with consciousness’ light seems to echo abrahamic ideas, with the twist (according to moynihan) that now we realise *we are alone*, so it adds a certain urgency (which is clear in bostrom etc.)
- this book by [Voeglin](#) might be interesting - ref’d in Moynihan - might talk about early church, katechism/eschaton, and has 2 chapters on gnosticism

- good quote (in an essay quite useful on the *genealogy of perfectability* from augustine, anselm, aquinas, duns scotus) from pete [wolfendale(<https://deontologistics.co/2019/10/22/tfe-immanentizing-the-eschaton/>)] (on *theology and perfectability*): “the idea that there is always an optimal solution to any problem, even if we cannot find it; whereas, in truth, perfectibility is a vanishingly rare thing. Using the term ‘perfectibility’ makes the connection to theology much clearer, insofar as it is precisely this that forms the analogical bridge between creator and created in the Christian tradition. Divinity is always conceptually liminal, and perfection is a popular limit.”
- AI in fiction might be useful somewhere: [wiki](#)

moynihan on peirce etc

- moynihan has an interesting hint at land’s argument against paperclippers, and negarestani’s notion of *intelligence/values as cultural/social thing constantly remaking itself* in this quote: ‘This is why, beyond the inarguable priority of ensuring existential security, it is perhaps misguided to construe alignment in a strong sense as the supergluing of synthetic mind to some contemporaneous picture of what humans are and value. Like all self-images proposed throughout the past, any such picture will, given time, undoubtedly be proven to have been a premature reification. Shackling mind to such a fleeting snapshot would be a betrayal of what’s been most valuable about humans so far: namely, that we are a work in progress.’
 - he cites [peirce](#) in the essay, abstract: “Knowledge a priori has an important role in rationalistic schools: it pre-establishes truth in order to justify a system of correlated ideas. Empiricism usually concerns knowledge a posteriori, for experience itself is what we have actually known. Peirce’s probabilistic approach to science was based on necessity in the long run but it has no clear place in the categorization of knowledge either as a priori or as a posteriori. This deficit should be overcome by introducing a new category — synthetic knowledge a ulteriori, defined as what is known about an indefinite number of cases but not about isolated instances.”

evidence from LW/EAF/AF

- has anyone actually done the research on religious inclinations of AI, silicon valley, or tech? Like another [Vox](#) (mostly about lemoine incident, pretty anecdotal pop psy stuff), this looks [possibly good](#)

or [Chen's work/pray/code](#) might be good for data

- probably need to somewhere define ‘apocalypse’ and ‘singularity’ as they are used in modern x-risk/rationalist circles, this [astralcodex](#) might be useful
- joe carslmith on religious takes on [the problem of evil](#)

A Faustian choice

The takeaway from the religious framing of AI x-risk is that current-day humans are faced with a problem often discussed in literature - that of Pandora’s Box or that faced by Dr. Faustus in Goethe’s eponymous work.

On one side are those that advocate technological advancement, whether as quickly as can be safely done (from an x-risk perspective), or accelerated without undue concern for hard-to-quantify risks (as the adherents of e/acc propose); against those that demand a halt to most or all research that might lead to TAI, many of whom are technically-literate AI researchers.¹⁹⁵ Some non-AI commentators, such as [Sacacas 2015](#) agree, and point out the importance of going slowly, not losing the world, imperfect as it is, that we currently have. He quotes [Noble 1999]:

“The religion of technology, in the end, ‘rests on extravagant hopes which are only meaningful in the context of transcendent belief in a religious God, hopes for a total salvation which technology cannot fulfill By striving for the impossible, [we] run the risk of destroying the good life that is possible.’ Put simply, the technological pursuit of salvation has become a threat to our survival.”

Sacacas cites, more as a conundrum facing humans, without indicating approval or disapproval, a passage from Wendell Berry:

“To recover from our disease of limitlessness, we will have to give up the idea that we have a right to be godlike animals, that we are potentially omniscient and omnipotent, ready to discover ‘the secret of the universe.’ We will have to start over, with a different and much older premise:

¹⁹⁵For one example, see this (very) short statement from the [Center for AI Safety](#) signed by most of the leading AI researchers. *This needs to be restated, rather than 2 camps, it's a bunch of different proposals. Even the ethics crowd (DAIR, etc) might not actively push for shutting down all research. We haven't reached the luddite/butlerian jihad moment yet.*

the naturalness and, for creatures of limited intelligence, the necessity, of limits. We must learn again to ask how we can make the most of what we are, what we have, what we have been given.”

These are philosophies of limits, the recognition of inaccessible mysteries, something that, at least from classical Greek thought, has been suggested, but which, as [Moynihan 2023] points out (“A split between those who, on one hand, respond to this promised climax with profound wariness and those who, on the other, express an impatient desire to usher in its arrival.”), was lost in the transition from Christian theology to science.^{196 197}

Random paras

However, I am interested in the TAI-specific context, which has been noted by [Moynihan 2023] and [Wolfendale 2019]. To recap, the former suggests that ideas of creating utopia-on-Earth that are specific to Christian doctrine (particularly as it became clear that salvation was not, in fact, at hand), might have left conceptual substructures or “desires”, that taint current thinking on AI (both in the potential some people see, and the risks others perceive, with AI; it is worth noting that these sets are not [disjunct]). Moynihan suggests that rather than seeking or forcing a solution, we should recognise that the future will be created in its own time, and the problems we perceive now are most unlikely to be solved through anything we create today (or in the future), including TAI.

[Wolfendale 2019] takes a different tack: he focuses on the (recursive) self-improvement and optimisation aspects of the AI x-risk narrative (to be discussed in detail in a later chapter, on the intellectual history of AI alignment), noting the similarities and genealogy from the perfectionability ideas in Christian and [Medieval Scholastic] writing. Wolfendale finds these ideas/fears reductive and un-grounded: they misunderstand reason, over-index on evolutionary analogies, and seem to miss the lessons from the revolutions of early twentieth-century mathematics, such as incompleteness and halting problems.¹⁹⁸

teilhard de chardin and huxley (see moynihan’s book). moynihan’s piece is really useful, esp. on teilhard

¹⁹⁶*Find Noble. Also, in relation, this might be useful on defn of [millenarianism](#) though note that is different from millenarianism*

¹⁹⁷*John Gray’s book also seems to have a distinct take on this, that christian/gnostic/enlightenment ideas have landed us in a weird place, and we’d do better to consider pre-christian or early judaism. This feels quite fertile. See meghan o’gieblin’s review of Gray for boston review (cited above) and the book has been ordered.*

¹⁹⁸*Try finding source in revenge of reason, but if all else fails, the gradient podcast 57:00 to 1:13:00 or so. Also wolfendale 2019 merits a reread after the podcast.*

-quote: 'Writing several years after his moonlit epiphany, Teilhard admitted that intelligence's apotheosis, in his eyes, demands "evacuation, pure and simple, of the old mankind." It can only be consummated in our "death." He compared the arrival of the posthuman to a tumor: budding off from humanity, "pushing it aside and taking its place."

As Peter Wolfendale puts it:

"There are plenty of people who have already pointed out the theological valence of the conclusions reached on the basis of these assumptions (e.g., the singularity, Roko's Basilisk, etc.); but these criticisms are low hanging fruit, most often picked by casual anti-tech hacks."
([Wolfendale 2019](#))

Studies indicate that people working on AI are relatively highly-educated, often male, and non-U.S. born, though their religious affiliations are less obvious.¹⁹⁹ So to assess the supportability of this narrative, we need to look deeper into possible histories that might plausibly influence what AI researchers think, for which there is only occasional evidence, like the post (quoted in [Samuel 2023]) on X from Anthropic's policy head Jack Clark, where he states:

"Sometimes I think a lot of the breathless enthusiasm for AGI is misplaced religious impulses from people brought up in a secular culture. When I was a kid I read a lot of religious texts of my own volition (and I continue to do so). For thousands of years, people have wanted to understand/be closer to some theistic god. There's some lesson to this re: the current AI boom that I don't quite yet understand." ([Jack Clark](#)).

Nick Land/acc stuff

Imagined dialogue between Land and an alignment researcher

Readers of this section might benefit from an understanding of current issues in the alignment of AI systems to human survival as a biological species.²⁰⁰

¹⁹⁹See [Abdulla 2023](#) for a demographic analysis of cited AI researchers, though it doesn't include race information. See below for some relevant statistics, albeit for a subset of AI-interested people e.g. self-identifying or associating members of fora devoted to rationalism, AI alignment, and effective altruism.

²⁰⁰Maybe should be moved to alignment chapter

There are a numbers of responses that a “modal” alignment researcher²⁰¹ might give in response to Land²⁰², but the main takeaways are as follows:

- Land does engage directly with Bostrom’s orthogonality and Omohundro’s instrumental drives ([Land 2023], pp. 137-138), but he seems to [misunderstand/mis-state] Bostrom’s point as “values are transcendent in relation to intelligence”.²⁰³ Land basically accepts Omohundro’s instrumental drives and uses it as support for Land’s position that humans don’t have a future in an AI-dominated world, but again his argument hinges entirely on a theoretical understanding of “intelligence”. This isn’t hugely controversial, in that he is just restating the Second Law of Thermodynamics, but he uses this linkage between problem-solving and (local) entropy increase in various ways, including a) loosely justifying his interest in time loops, and b) his critique of Kantian universalism e.g. in *Against Universalism* and more problematic polemics like *Economic Ends* or *Premises of Neoreaction*.] without engaging with concrete research.²⁰⁴
- The words “superintelligence” and “intelligence” do a lot of work in Land’s writing, as they do for Yudkowsky, Bostrom, *et al.* However, of the three, Land does the least to rigorously define and stress his usage of the terms, though similar critiques have been leveled at the others (usually by empirical AI risk researchers). Land speaks of superintelligence at such a high level of abstraction that he can’t engage with detailed arguments set out by various writers on why even near-current level AIs could have severely damaging, in the sense of destroying civilisation or the markets that are Land’s primary historical force, impacts (therefore possibly never even getting to the stage of superintelligence). Other examples are:
 - For instance, his definition of intelligence as “game-playing competence” ([TU 2024] 17:00), is, at least, inadequately specified (what exactly is a “game”?) - which feels vaguely unsatisfactory for such a load-bearing term. In the podcast, he seems to define the term “game” in maximal terms (i.e. everything humans do), rendering it potentially unarguable or unanalysable.²⁰⁵ He

²⁰¹Say, one that was a compressed version of Paul Christiano, Eliezer Yudkowsky, Matthew Barnett, Katja Grace, Ajeya Cotra, Evan Hubinger, Daniel Kokotajlo, Janus, Buck Shlegeris, Rohin Shah, Beth Barnes, Joscha Bach, Andrew Critch. This list is subjective, but I have used the following criteria (based on alignmentforum.org rankings): personal familiarity with views; gender balance (4.5/13 are not, to my knowledge, male-identifying); a mix of technical and philosophical perspectives; volume of output and community scoring/commenting.

²⁰²Transcripts are attached in the appendices.

²⁰³Go back to Bostrom/expand that section (in alignment chapter) as it is important and adjudicate this

²⁰⁴Tighten this up, what exactly is my critique of Land?

²⁰⁵Get the quote

writes more on “intelligence” in *What is Intelligence?*, as well as *Extropy* in [Land 2023], the gist of which is that he thinks of it as problem-solving capacity and the creation of information (not defined in any specific or Shannon-esque way), and critically, with a *local increase in entropy* which Land terms “extropy”.²⁰⁶

- * Some other (alignment-discourse relevant) conceptions of intelligence²⁰⁷ include: optimisation power (the ability to hit a small target in a large search space); intelligence as building/updating world-models that minimise predictive error; ability to generate abstract patterns and manipulate them (not necessarily with any goal) and ability to transfer abstractions across domains; ability to learn and satisfy complex preferences that don’t easily correspond to concrete objectives; ability to reason about counterfactual worlds; ability to integrate information across scales and modalities; ability to explain phenomena through minimal descriptions (e.g. compression); ability to meta-learn (learn how to learn, which is more about process than outcome); ability to manipulate causal models²⁰⁸
- Land does not claim to engage with developments in deep learning, RL, or even LLMs, and his only references to the AI literature are quite specific. Besides Yudkowsky, Bostrom and Omohundro (all in [Land 2023] even though they were quite active as of [Land 2011], and known to the editors of [Mackay 2014] as we see in [*Collapse V*]), his references, while seminal, are early in the discourse: e.g. mathematician I.J. Good, founder of cybernetics Norbert Wiener, information theorist Claude Shannon, the fiction of Samuel Butler; there is also an undercurrent of writers on self-reference like Gödel and Hilbert; and of course a broad range of continental philosophers. This is a problem in the sense that AI (circa 2024) is an empirical discipline, more so than it was in 2011-2014. For instance
 - * Researchers seem to be confused as to how human-given goals are actually represented within a model (the inner alignment or mesa-optimisation problem);
 - * It is unclear whether model actually believe something (in the sense of thinking it is “true” by some measure) or is it pretending or predicting what a human evaluator would like to

²⁰⁶Entropy is referenced in [Land 2023] much more frequently, in various guises, some quite politically problematic, at least from a centrist/left-coded perspective. Entropy production-as-intelligence (in *On Chaos*) has in turn been picked up by the e/acc movement, adding to its vaguely air of absurdity.

²⁰⁷From Claude-3.5-sonnet-new

²⁰⁸Look in sonnet transcript for names, might need further explication/references perhaps from LW/PIBBSS reading

- hear (the Simulators framing or the prediction problem, respectively);
- * Are all AIs necessarily goal-directed or agentic (again, the Simulators framing and contemporary work on scaffolded agents);
 - * Might near-current models learn human (some approximation of some subset of) values through pretraining and post-training processes like RLHF (which actively try to shape the model's responses towards things human evaluators rate as "good"), so that alignment actually is easier than originally supposed (in the MIRI View)?²⁰⁹
- Land ignores the path-dependency of AI development: his ultimate descriptive vision is of superintelligence edging out the human, while (my presumption is) undertaking broadly scoped goals that are in a sense "worthy". He dismisses by fiat²¹⁰ the possibility of a paperclipper (and presumably anything roughly similar) by combining volition and intelligence in one bag-to-be-optimised, without spelling out a vision of what such a superintelligence would actually *do*. In fact, in keeping with his descriptive rather than normative focus, Land doesn't in [Land 2011], [Mackay 2014], [Land 2023], [TU 2024] make statements as to what sort of human succession would be "valuable" (even by the "lights" of Gnon, his avatar for the will-of-nature, something like Bostrom's "cosmic host"²¹¹), putting him into the camp of radical cosmopolitanism (as Christiano describes it)²¹²
- Insofar as markets are doing anything describable as optimisation, they seem to be doing this locally, rather than at any global or broad scale, and there doesn't seem to be any goal or overall target towards what markets are moving (any more than evolution, which seems to operate towards a very local purpose, namely genetic fitness across a single generation, isn't normally thought of as having a trans-generational goal or optimisation target)
 - Land's critique of the history of feedback in cybernetic or economic systems as being dominated by negative, homeostatic feedback is insufficiently detailed:
 - positive feedback loops are inherently destabilising, and can lead to system collapse (e.g. banks that become too-large-to-fail or species that multiply so much that they exhaust their food sup-

²⁰⁹Recap joe carlsmith arguments on exactly this in dwarkesh podcast

²¹⁰what is the philosophical word for this...dismisses

²¹¹These topics will be explored in a future chapter on the point-of-view of the universe.

²¹²Check the christiano cosmopolitanism which comes from EY probably, fragility of value , etc

ply);

- pure positive feedback loops tend to be locally and temporally limited for this reason, as physical limits are hit.
- in our reality, evolutionary and human-made systems often involve complex balancing of positive and negative feedback (life itself, in some conceptions²¹³ or government regulation of companies).
- selection bias means we mainly see systems that have survived, though the historical record shows species, civilisations and companies that collapsed for reasons traceable in part to unchecked growth.
- however, Land’s emphasis on positive feedback holds some weight in that to the extent new resources can be acquired, a given entity can continue to grow for a very long time indeed. Land sees this in an evolutionary/thermodynamic frame, seeing life, as well as civilisational artefacts such as technology and markets, as local decrease in entropy within the boundaries of the organism or system (albeit with overall entropy increase, usually dumped as waste heat or products to some reservoir outside the organism/system). As he says in [TU 2024], this process can go on for a very long time.

* This is good as far as it goes. But he doesn’t quite explain, in AI relevant terms, how a maximally powerful optimising system (to borrow from Omohundro, which Land seems broadly in agreement with) could accidentally destroy much of the highly-structured features of our world (e.g. life and civilisation) without actually replacing it with anything of comparable complexity. He just excludes this possibility²¹⁴.

* He doesn’t even note the possibility of mistakes - between, say, human-level AI and AGI, and from there to superintelligence, there seem like many opportunities to get things wrong, resulting in vast loss of complexity (increases in entropy), without an obvious path for the superintelligence to “rise from the ashes”.²¹⁵

- Land’s teleological approach seems to imply that human actors (e.g. in governments, regulators, corporations, as well as scientists) have no substantial ability to influence the course of the near future,

²¹³Cite

²¹⁴Quote

²¹⁵Cite where this is found in his writing, but also is core of e/acc and worth explaining how entropy/negentropy thing works

and the forces of runaway market-based optimisation would lead to scenarios that disempower humans. This has some apparent similarity to the writing around “Moloch” and general international and corporate coordination failures, however Land is going further by seeming to impart agency to the system itself.

- in response, GPT-4o (acting in the persona of Land) says:²¹⁶

²¹⁶Move to transcript

Glossary

AI, ML, DL, AGI, ASI

Start with this [kate crawford](#) reference

Acausal Decision Theories

Alignment

Pre-training

RLHF

Coherent Extrapolated Volition (CEV)

Human Level Machine Intelligence (HLMI)

Timeline of AI Capabilities and Alignment Progress

Date	Item	Significance	Link
1943	“A Logical Calculus of the Ideas Immanent in Nervous Activity” by McCulloch and Pitts	Proposed the first mathematical model of a neural network, laying groundwork for future AI research	https://link.springer.com/article/10.1007/BF02478259
1944	“Theory of Games and Economic Behavior” by von Neumann and Morgenstern	Laid the foundation for game theory, influencing AI approaches to decision-making	https://press.princeton.edu/books/hardcover/9780691130613/theory-of-games-and-economic-behavior
1950	“Computing Machinery and Intelligence” by Alan Turing	Introduced the Turing test, a fundamental concept in evaluating machine intelligence	https://academic.oup.com/mind/article/LIX/236/433/986238
1951	First neural network machine (SNARC) by Marvin Minsky	Built the first neural network simulator, demonstrating early hardware implementation of AI concepts	https://web.media.mit.edu/~minsky/papers/SNARCMachine.html
1956	Dartmouth Workshop	Coined the term “artificial intelligence” and marked the birth of AI as a field	https://www.dartmouth.edu/~ai50/history.html
1956	Logic Theorist by Allen Newell, Herbert A. Simon, and Cliff Shaw	First program deliberately designed to perform automated reasoning, pioneering symbolic AI	https://history-computer.com/logic-theorist/
1958	Perceptron by Frank Rosenblatt	First artificial neural network, demonstrating early machine learning capabilities	https://www.cs.cmu.edu/~epxing/Class/10715/reading/Rosenblatt.pdf
1959	“Programs with Common Sense” by John McCarthy	Introduced the idea of a logical AI system, influencing symbolic AI approaches	http://jmc.stanford.edu/articles/mcc59/mcc59.pdf

Date	Item	Significance	Link
1965	“On the Nature of Human Thinking” by I.J. Good	Introduced the concept of an “intelligence explosion,” influencing later ideas on AI safety	https://academic.oup.com/mind/article-abstract/LXXIV/293/1/984316
1969	General Problem Solver (GPS) by Newell and Simon	Showcased problem-solving approaches in AI, a cornerstone of symbolic AI	https://www.sciencedirect.com/science/article/abs/pii/S0004370297000513
1972	PROLOG programming language	Developed for logical programming, becoming a staple of symbolic AI	https://dl.acm.org/doi/10.1145/359460.359470
1980	Expert systems gain prominence	Demonstrated practical applications of symbolic AI in various domains	https://ieeexplore.ieee.org/document/6312920
1986	Backpropagation algorithm popularized	Efficient method for training neural networks, crucial for later deep learning advances	https://www.nature.com/articles/323533a0
1992	Reinforcement Learning Breakthrough	Advancements in algorithms like Q-learning, contributing to autonomous decision-making	https://link.springer.com/article/10.1007/BF00992698
1997	Deep Blue defeats Garry Kasparov	IBM’s supercomputer beats world chess champion, demonstrating AI’s potential in strategic games	https://www.ibm.com/ibm/history/ibm100/us/en/icons/deepblue/
2012	AlexNet	Deep convolutional neural network that won the ImageNet challenge, triggering an explosion in deep learning research	https://papers.nips.cc/paper/2012/hash/c399862d3b9d6b76c8436e924a68c45b-Abstract.html
2014	Generative Adversarial Networks (GANs)	Introduced by Ian Goodfellow et al., enabling realistic AI-generated images and videos	https://arxiv.org/abs/1406.2661

Date	Item	Significance	Link
2015	“Concrete Problems in AI Safety” paper	Identified key challenges in ensuring AI systems behave safely and as intended	https://arxiv.org/abs/1606.06565
2016	AlphaGo defeats Lee Sedol	Google DeepMind’s AI beats world champion in Go, showcasing advanced strategic reasoning	https://deepmind.com/research/open-source/alphago
2016	Establishment of OpenAI	Founded with the mission to ensure artificial general intelligence benefits all of humanity	https://openai.com/about
2017	“Attention is All You Need” paper	Introduced the Transformer architecture, laying the foundation for large language models	https://arxiv.org/abs/1706.03762
2017	Asilomar AI Principles	Guidelines for beneficial AI development, agreed upon by leading AI researchers	https://futureoflife.org/open-letter/ai-principles/
2018	GPT-1 released by OpenAI	First Generative Pre-trained Transformer, marking a significant step in language models	https://openai.com/research/language-unsupervised
2018	AI Alignment Research Overview by 80,000 Hours	Comprehensive review of AI alignment challenges and research directions	https://80000hours.org/problem-profiles/artificial-intelligence/
2019	BERT by Google	Bidirectional Encoder Representations from Transformers, advancing natural language understanding	https://arxiv.org/abs/1810.04805
2019	Launch of the Center for Human-Compatible AI (CHAI)	Research institute focused on ensuring that AI systems are aligned with human values	https://humancompatible.ai/
2020	GPT-3 released by OpenAI	175 billion parameter language model, demonstrating unprecedented natural language capabilities	https://arxiv.org/abs/2005.14165
2021	DALL-E by OpenAI	Text-to-image generation model, showcasing advanced multimodal capabilities	https://openai.com/research/dall-e

Date	Item	Significance	Link
2021	“On the Opportunities and Risks of Foundation Models” paper	Comprehensive analysis of the impact of large-scale AI models on society	https://arxiv.org/abs/2108.07258
2022	ChatGPT released by OpenAI	Conversational AI model that gained widespread public attention and adoption	https://openai.com/blog/chatgpt
2022	“Constitutional AI” paper by Anthropic	Proposed approach for aligning AI systems with human values and principles	https://arxiv.org/abs/2212.08073
2023	GPT-4 released by OpenAI	Advanced multimodal model with improved reasoning and task completion abilities	https://openai.com/research/gpt-4
2023	“Statement on AI Risk”	Signed by prominent AI researchers and tech leaders, calling for prioritizing AI safety alongside global risks	https://www.safe.ai/statement-on-ai-risk
2024	Claude 3 model family released by Anthropic	Advanced AI models focusing on improved capabilities and safety features	https://www.anthropic.com